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A GROUP OF B7H3 MONOCLONAL ANTIBODIES AND MEDICAL USE THEREOF

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Applicant: Innolake Biopharma (Hangzhou) Co., Ltd., Hangzhou (CN)

(72)

Inventors: Junzhuan QIU, Hangzhou (CN); Zhensheng WANG, Hangzhou (CN); Kai SUN, Hangzhou (CN); Junyong CHEN, Hangzhou (CN); Jian SUN, Hangzhou (CN); Zhongliang LI, Hangzhou (CN); Mingde XIA, Hangzhou (CN); Rulei CHEN, Hangzhou (CN)

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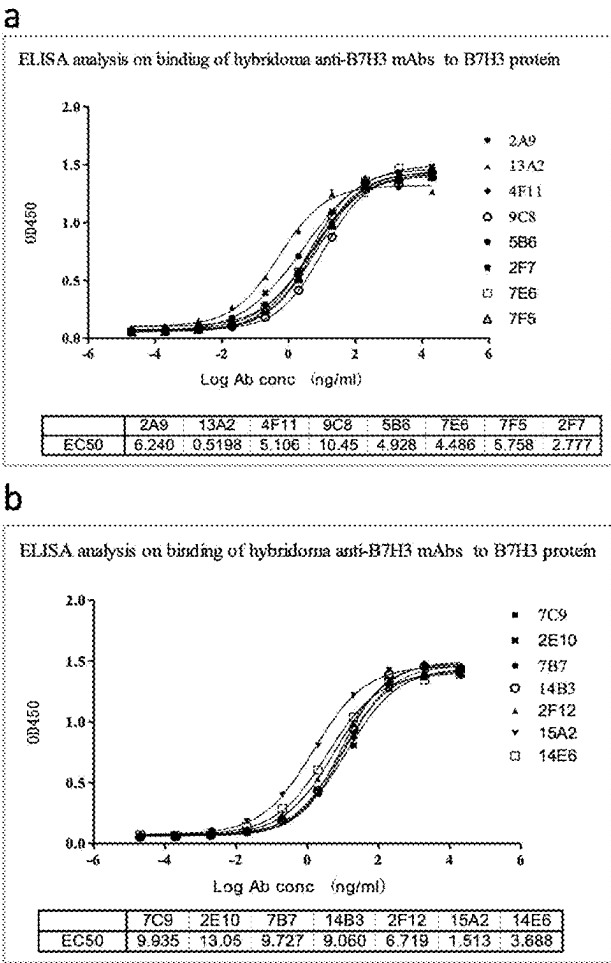
CPC A61K 47/6849 (2017.08); A61K 47/68031 (2023.08); C07K 16/2827 (2013.01); C07K 2317/24 (2013.01); C07K 2317/732 (2013.01); C07K 2317/77 (2013.01); C07K 2317/92 (2013.01)

(57)

ABSTRACT

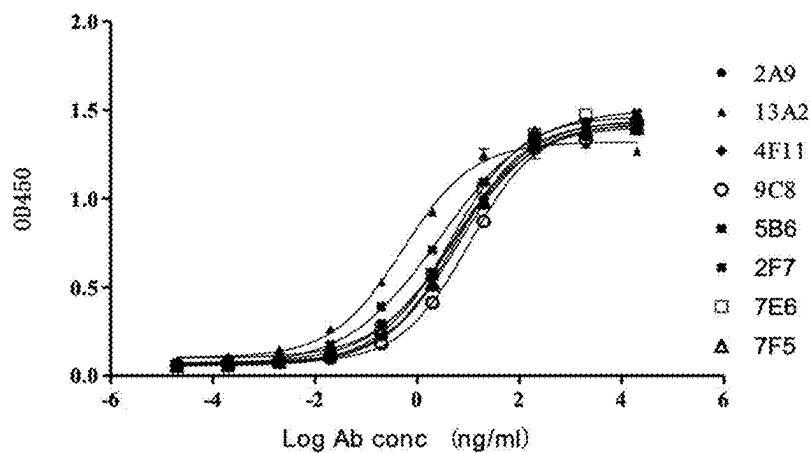
Provided are a group of anti-B7H3 monoclonal antibodies and humanized antibodies thereof. The antibodies obtained by means of hybridoma technology have B7H3 affinity, cell endocytosis effect, and antibody-dependent cytotoxicity, or can effectively block an immunosuppressive effect caused by B7H3, and can be used for preparing related drugs for inhibiting cancer cells, regulating the effect and level of B7H3, and enhancing the immunity of an organism.

Specification includes a Sequence Listing.



a

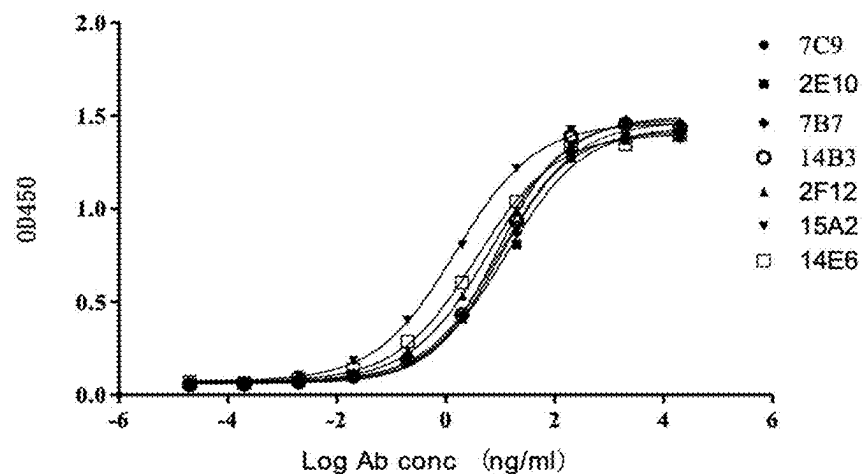
ELISA analysis on binding of hybridoma anti-B7H3 mAbs to B7H3 protein



	2A9	13A2	4F11	9C8	5B6	7E6	7F5	2F7
EC50	6.240	0.5198	5.106	10.45	4.928	4.486	5.758	2.777

b

ELISA analysis on binding of hybridoma anti-B7H3 mAbs to B7H3 protein



	7C9	2E10	7B7	14B3	2F12	15A2	14E6
EC50	9.935	13.05	9.727	9.060	6.719	1.513	3.688

FIG. 1

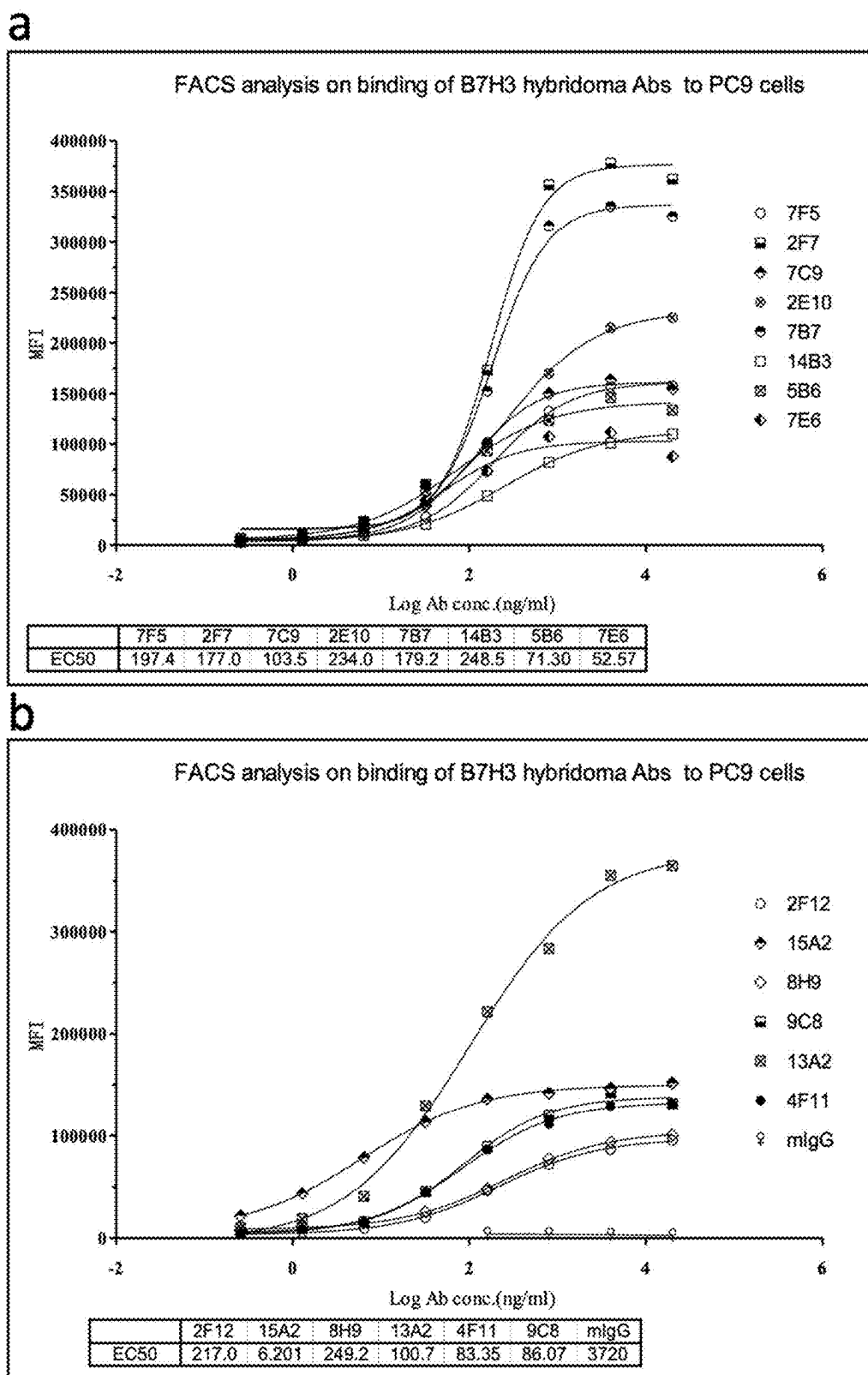


FIG. 2

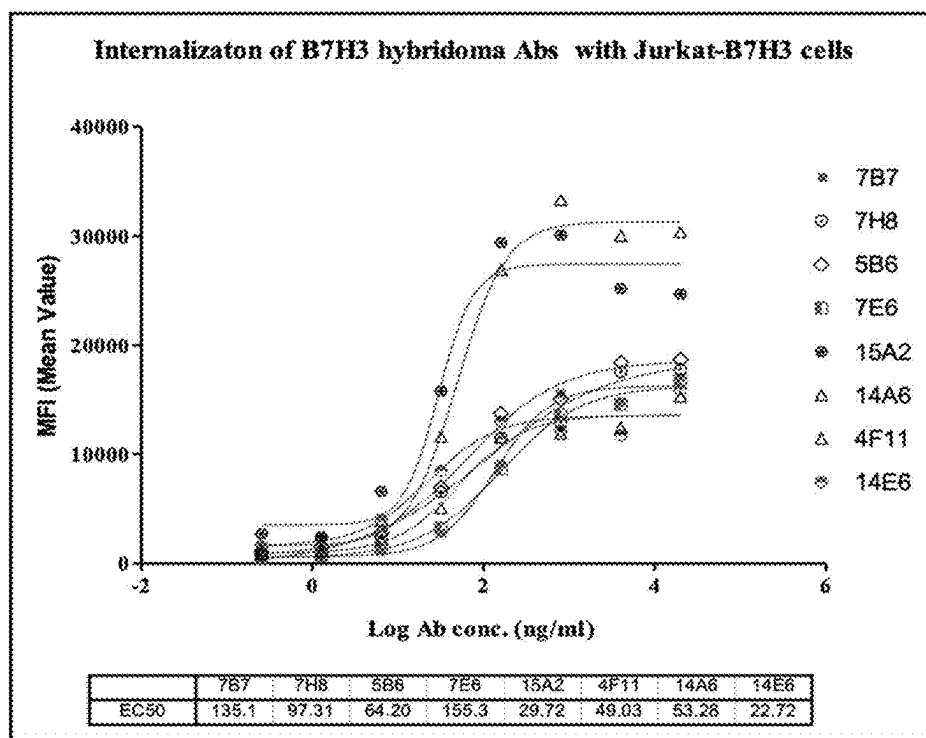
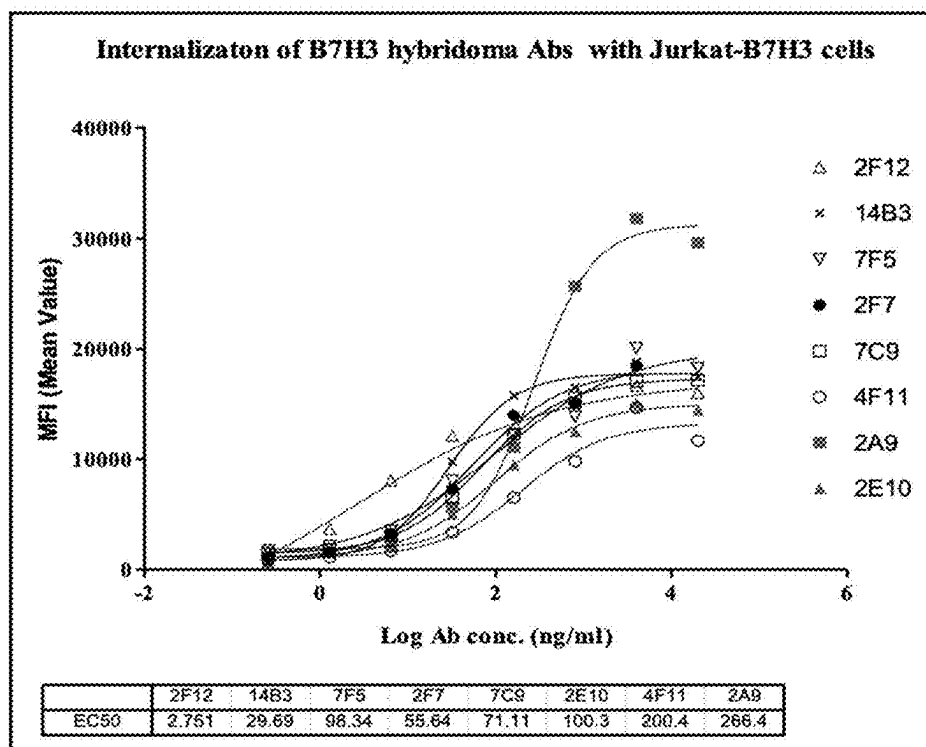
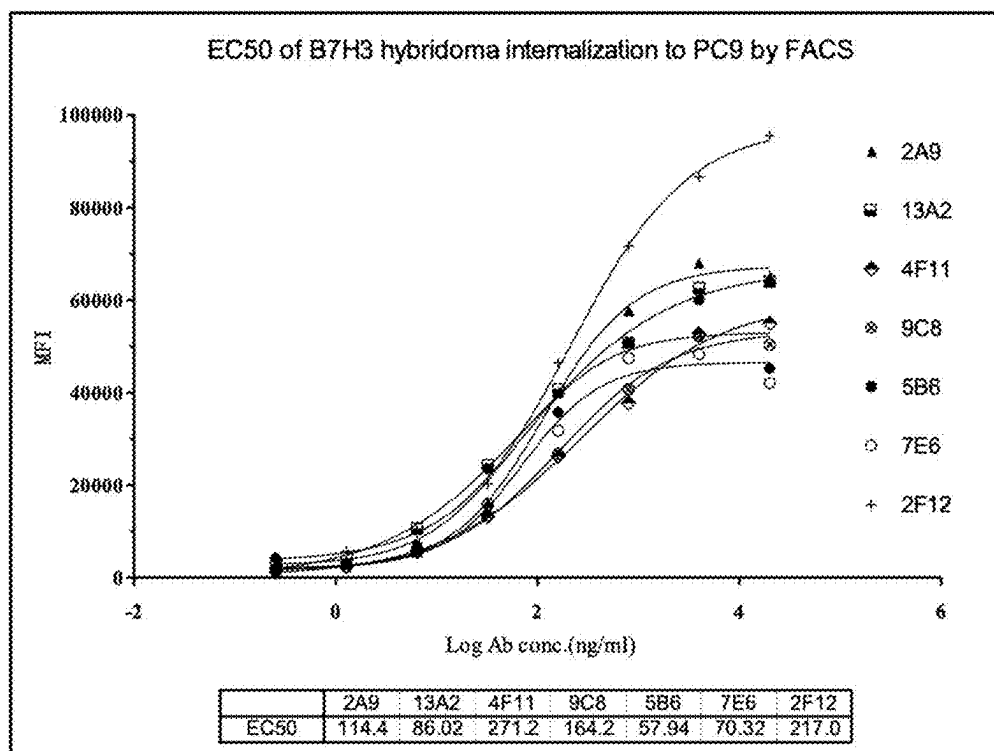
a**b**

FIG. 3

a



b

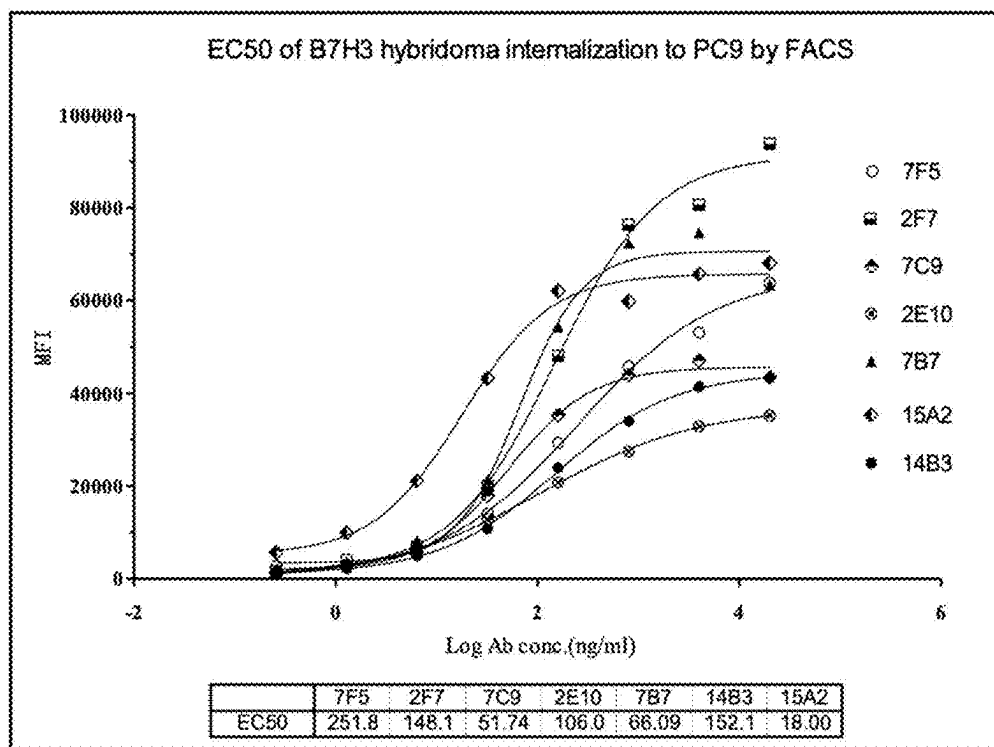
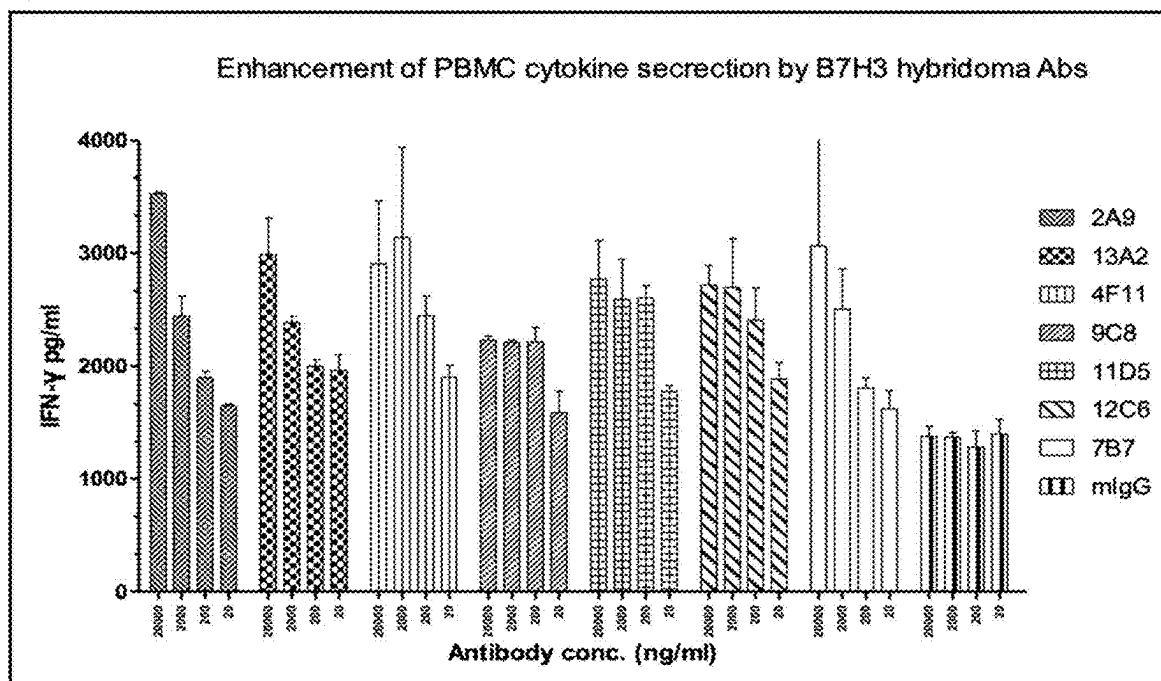


FIG. 4

a



b

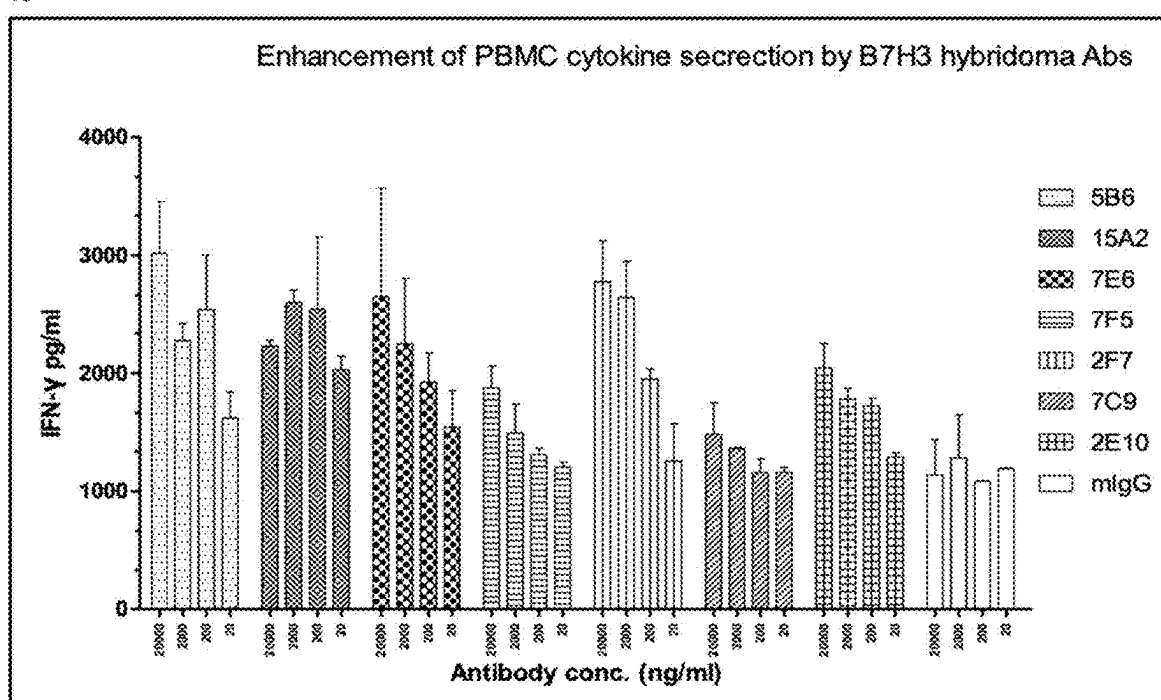
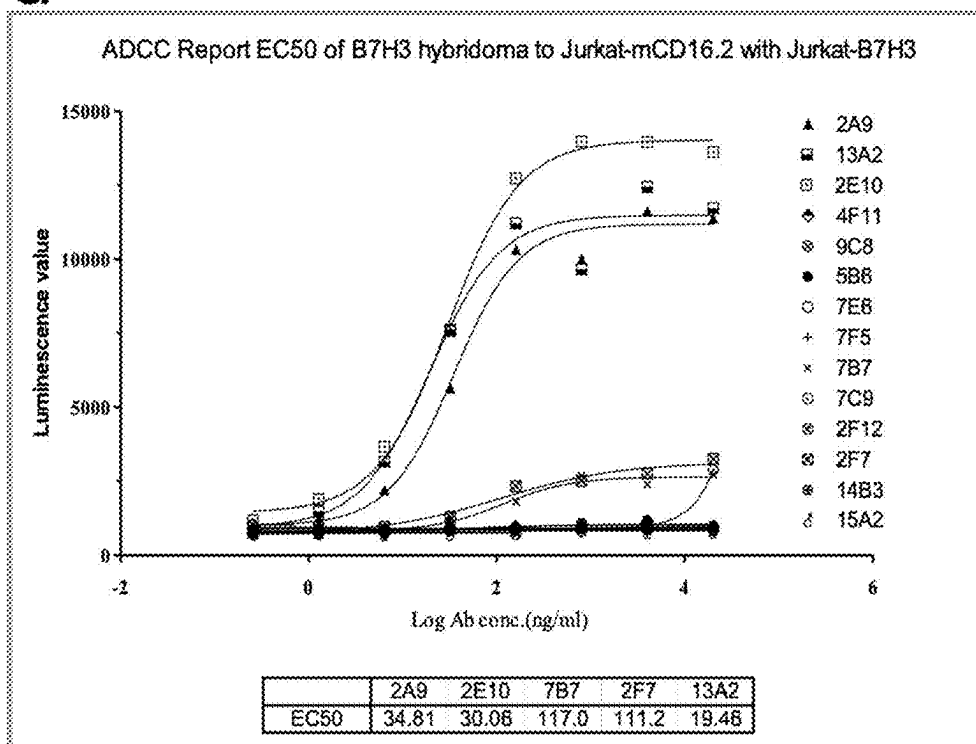


FIG. 5

a



b

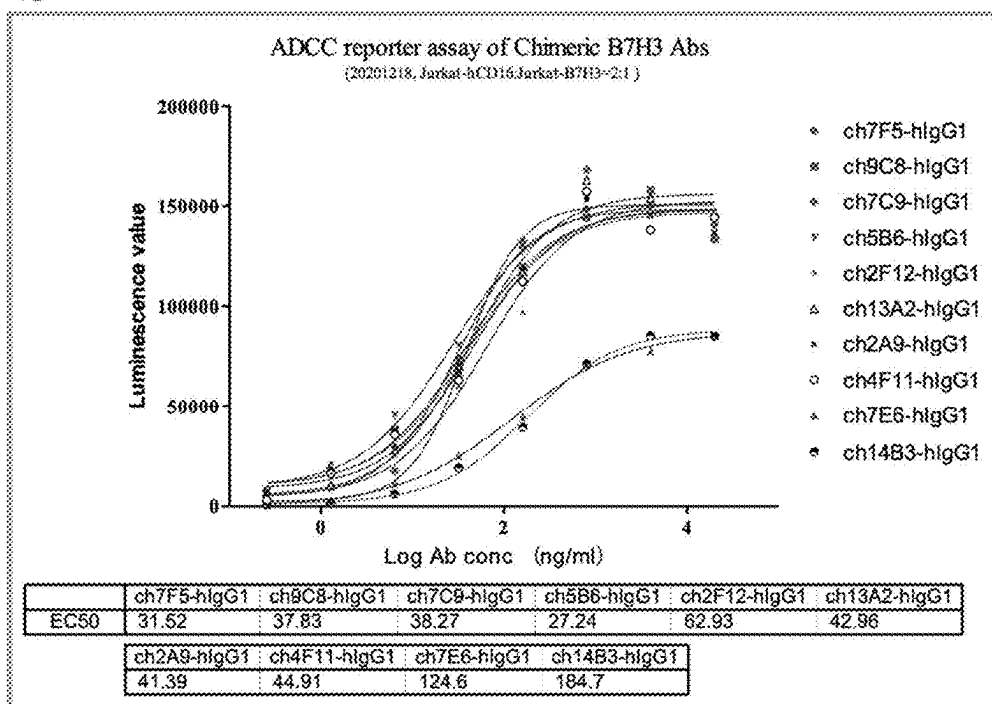
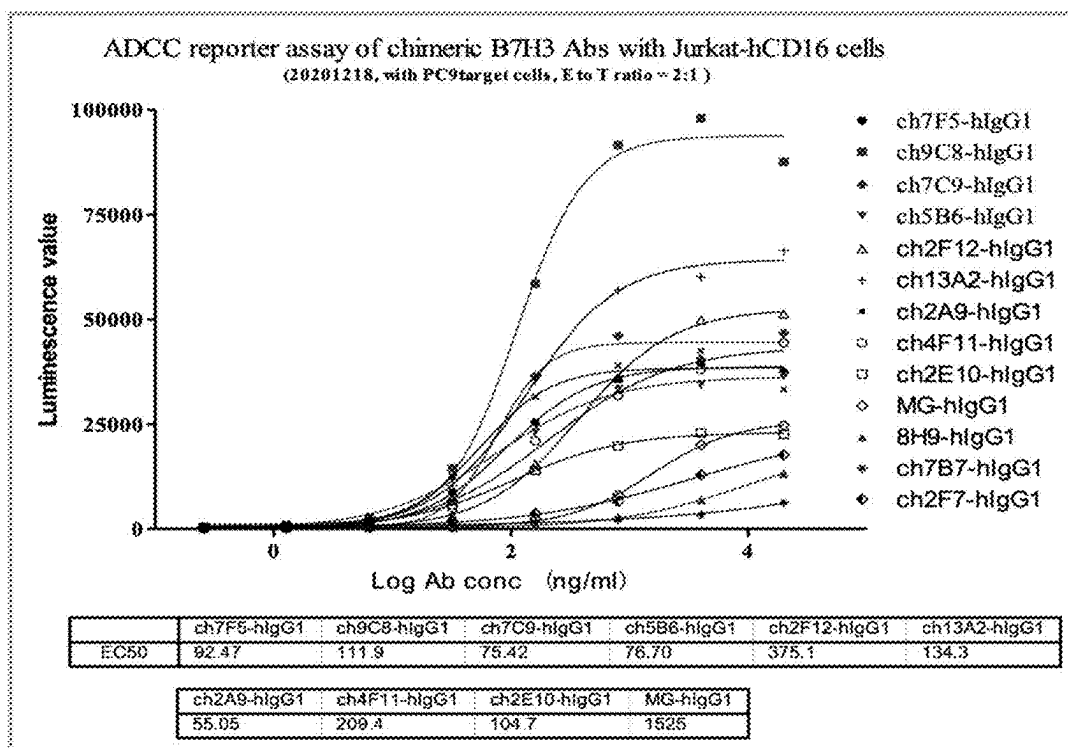


FIG. 6

a



b

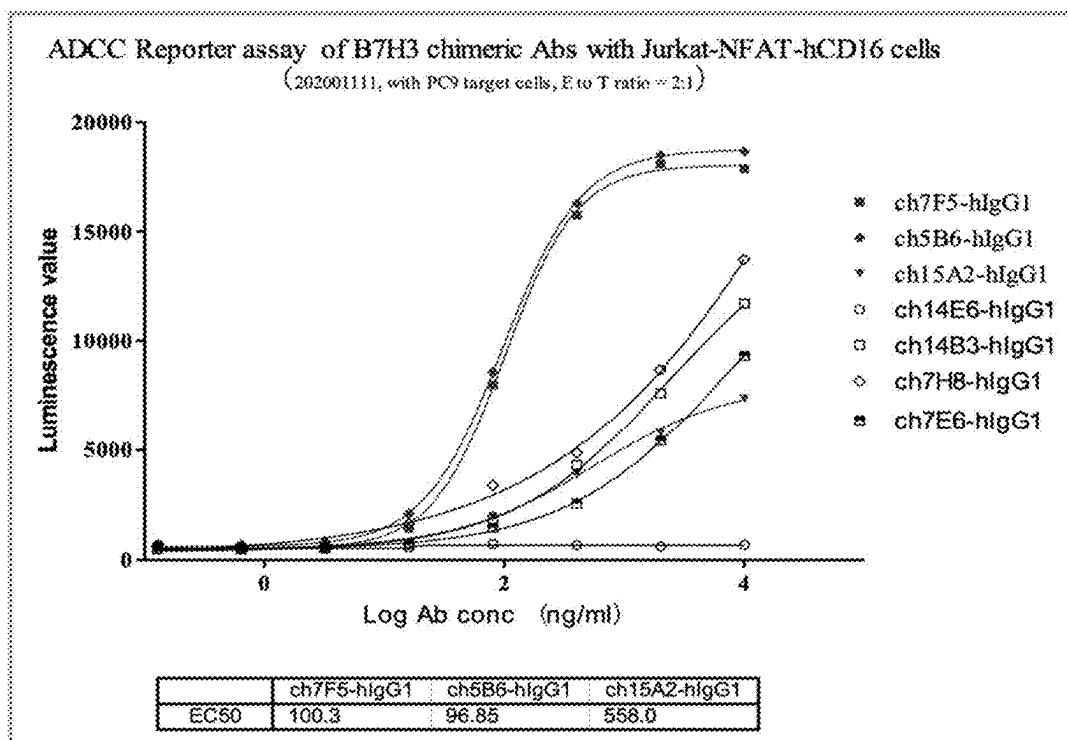
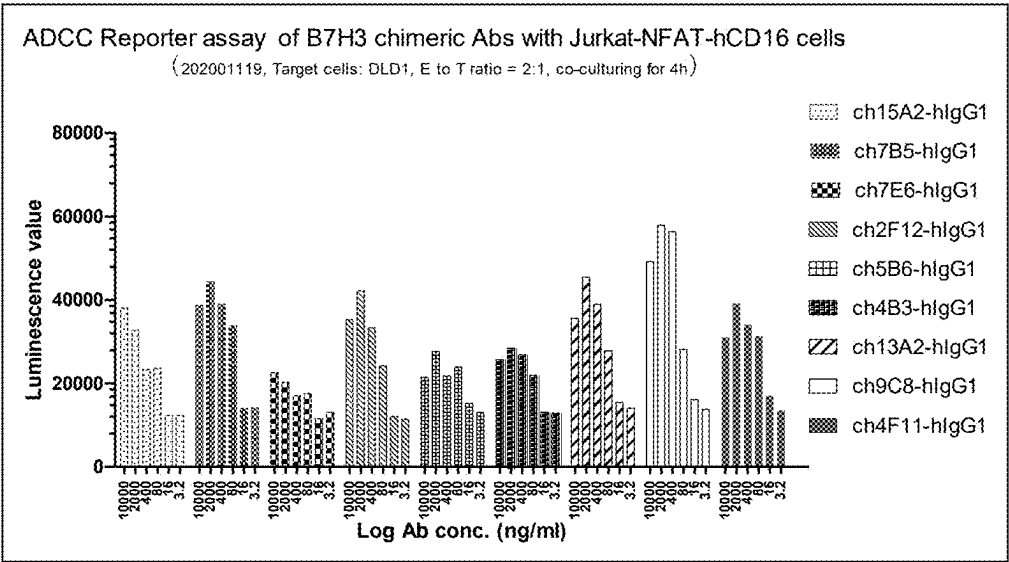


FIG. 7



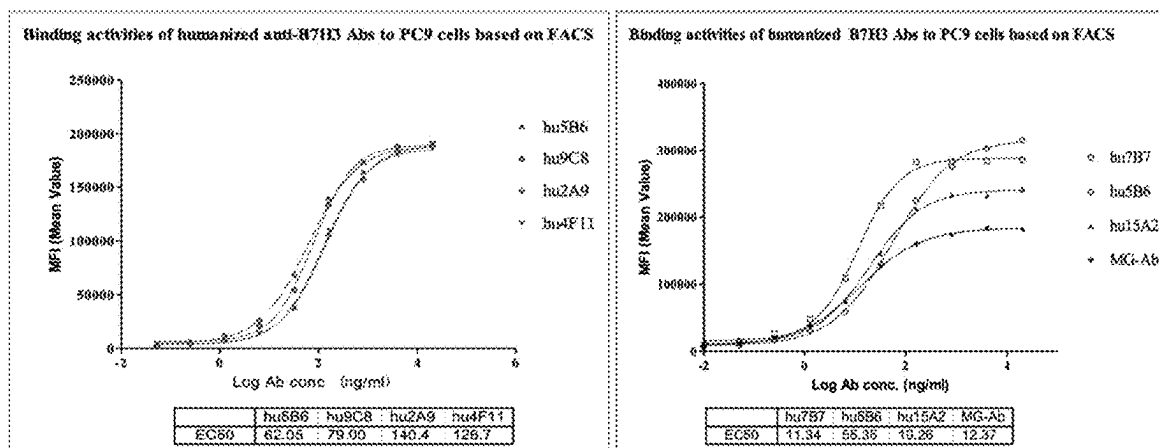


FIG. 10

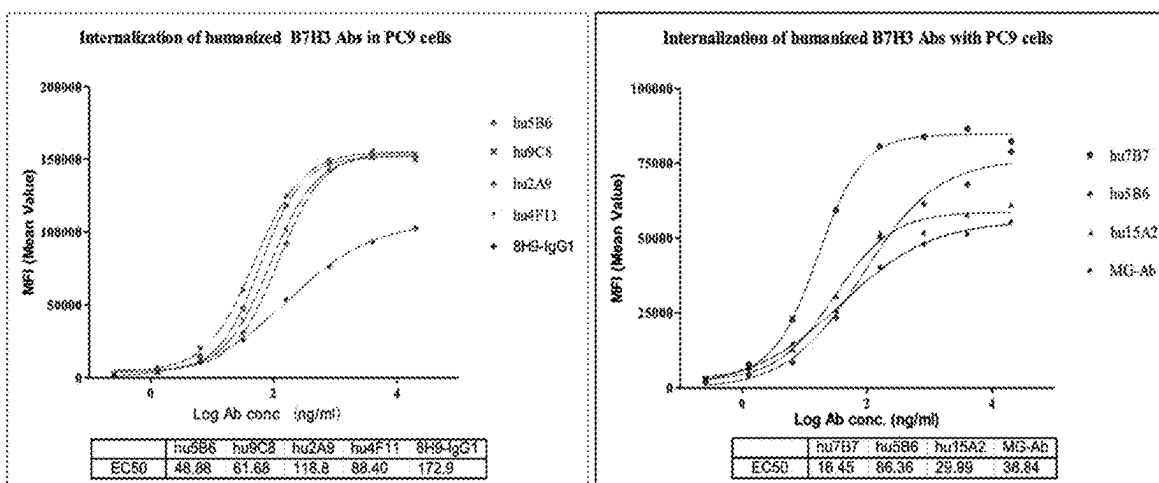


FIG. 11

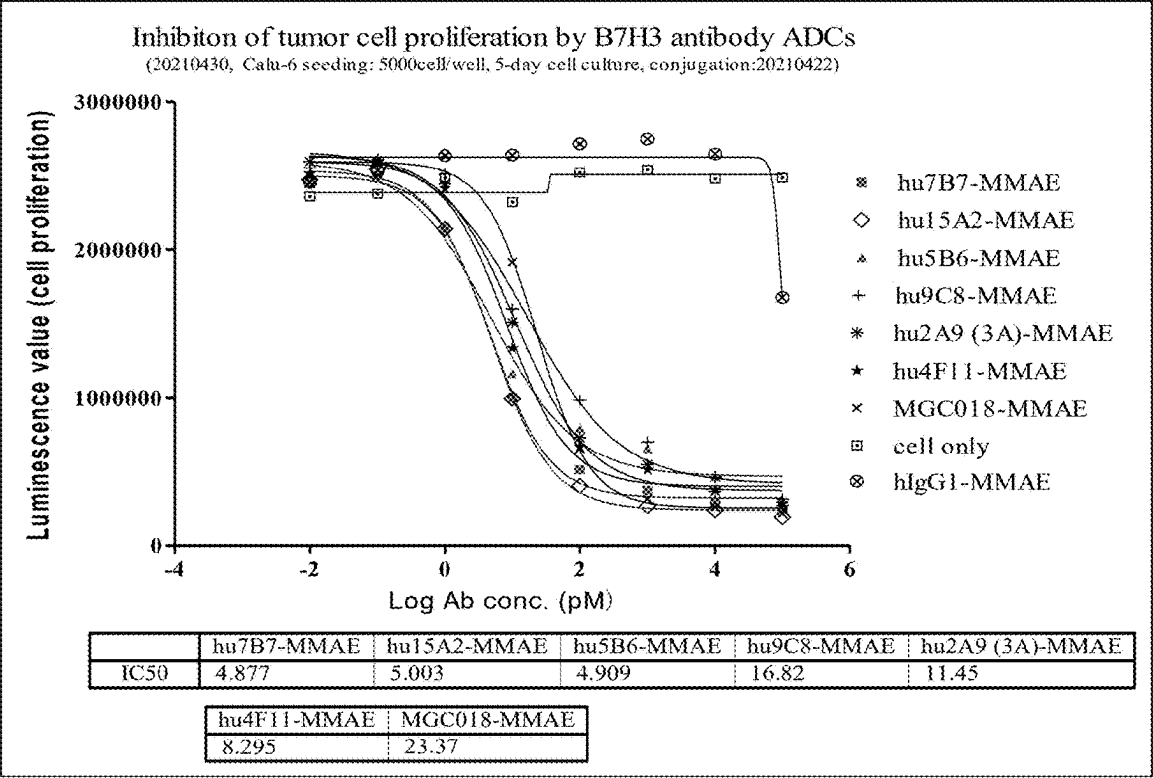


FIG. 12

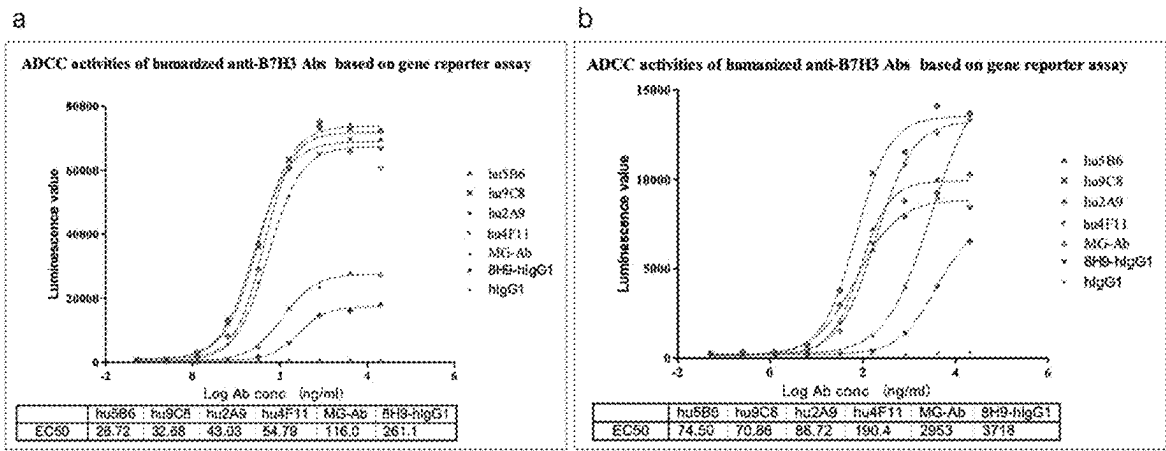


FIG. 13

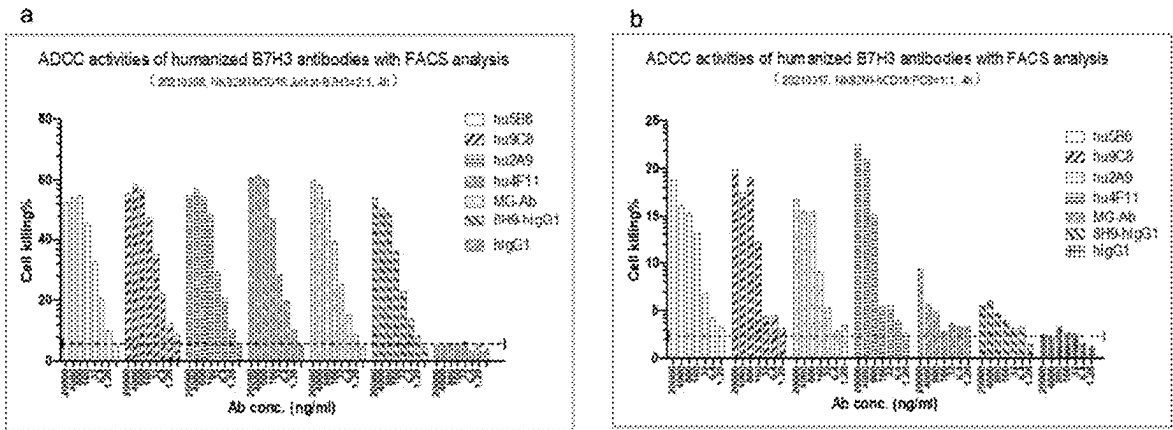


FIG. 14

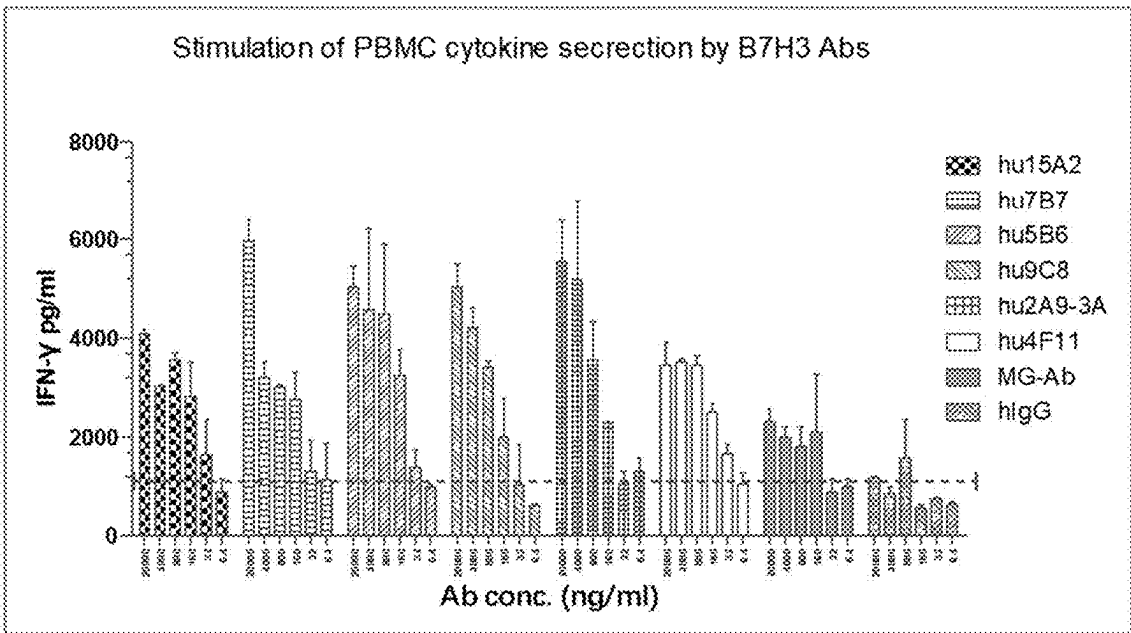


FIG. 15

A GROUP OF B7H3 MONOCLONAL ANTIBODIES AND MEDICAL USE THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202110639996.0, filed with the China National Intellectual Property Administration on Jun. 9, 2021 and entitled “A GROUP OF B7H3 MONOCLONAL ANTIBODIES AND MEDICAL USE THEREOF”, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure belongs to the field of anti-tumor antibody treatment and molecular immunology treatment, and relates to anti-B7H3 antibodies and use thereof. Specifically, the present disclosure relates to various anti-B7H3 monoclonal antibodies.

BACKGROUND

[0003] B7H3, also known as CD276, is a type I transmembrane protein (Chapoval A. I., et al., (2001), *Nat. Immunol.* 2:269). In mice, B7H3 is encoded by chromosome 9, while in humans, it is encoded by chromosome 15. B7H3 shares 20%-27% amino acid homology with other B7 family ligands (Sun M., et al., (2002), *J. Immunol.* 168:6294; Loos M., (2010), *Clin. Dev. Immunol.* 2010:683875). In terms of protein composition and structure, murine B7H3 has only two IgB7H3 types (consisting of a pair of extracellular IgV and IgC), and human B7H3 has four IgB7H3 types, which are almost identical tandem repeats of IgV-IgC (Sun M., et al., (2002), *J. Immunol.* 168:6294; Steinberger P., et al., (2004), *J. Immunol.* 172:2352). Although B7H3 mRNA is present in some normal tissues such as liver, small intestine, pancreas, testis, heart and colon, B7H3 protein is rarer in normal tissues (Greenwald R. J., et al., (2005), *Annu. Rev. Immunol.* 23:515-48). This difference between B7H3 mRNA and protein may reflect existence of close post-transcriptional regulatory mechanisms (Hofmeyer K. A., et al., (2008), *Proc. Natl. Acad. Sci.* 105:10277; Calabro L., et al., (2011), *J. Cell Physiol.* 226:2595). Generally, B7H3 is only present in some non-immune fibroblasts, endothelial cells, and osteoblasts, as well as some immune cells such as B cells, T cells, monocytes, dendritic cells, or NK cells. Expression of B7H3 can be induced by CSF or lipopolysaccharide stimulation in granulocyte-macrophage cells (Suh W. K., et al., (2003), *Nat. Immunol.* 4:899; Chapoval A. I., et al., (2001), *Nat. Immunol.* 2:269; Greenwald R. J. et al., (2005), *Annu. Rev. Immunol.* 23:515).

[0004] B7H3 belongs to important immune checkpoint proteins. Preliminary studies have shown that B7H3 has a stimulation effect on T cells. It can promote proliferation of CD4+ and CD8+ T cells to some extent, enhance the cytotoxicity of T lymphocytes, and stimulate production of interferon- γ (IFN- γ) (Chapoval A. I., et al., (2001), *Nat. Immunol.* 2:269). And deficiency of B7H3 can alleviate chronic rejection of cardiac allografts (Wang L. et al., (2005), *Eur. J. Immunol.* 35:428). However, more studies have shown that B7H3 also has significant immunosuppressive effects. For example, B7H3 can significantly inhibit activation effects of CD3 antibodies or allogeneic DC cells on T cells, and B7H3-blocking antibodies can effectively reverse the inhibitory effect (Prasad D. V. R., et al., (2004),

J. Immunol. 173:2500). Structural information of B7H3 protein reveals that FG loop in B7H3 IgV domain may have a critical effect on inhibition of T cells (Vigdorovich V., et al., (2013), *Structure* 21:707). In addition to T cells, B7H3 may also have an inhibitory effect on natural killer cells (NK) (Castriconi R., et al., (2004), *Proc. Natl. Acad. Sci.* 101:12640). A possible mechanism is that regulatory T cells (Tregs) induce the expression of B7H3 in dendritic cells (DC), thereby inhibiting a stimulatory effect of DC cells on T lymphocytes (Mahnke K., et al., (2007), *Eur. J. Immunol.* 37:2117). Mouse models lacking B7H3 exhibit more severe airway inflammations and earlier occurrence of some autoimmune diseases such as experimental autoimmune strain encephalomyelitis (Suh W. K., et al., (2003), *Nat. Immunol.* 4:899).

[0005] B7H3 is highly expressed in many malignant tumors such as melanoma (Wang J., et al., (2013), *J. Invest. Dermatol.* 133:2050), leukemia (Hu Y., et al., (2015), *Hematology* 20:187; Sun J., et al., (2014), *Oncotargets Ther.* 7:1979), prostate cancer (Zang X., et al., (2007), *Proc. Natl. Acad. Sci.* 104:19458), ovarian cancer (Zang X., et al., (2010), *Mod. Pathol.* 23:1104) and pancreatic cancer (Chen Y., et al., (2014), *Oncotargets Ther.* 7:1465-72). Although B7H3 mRNA exists in both normal and tumor tissues, B7H3 protein levels are higher in only tumor cells. This difference may be related to a regulatory mechanism of miRNA-29. For example, studies have shown that there is an inverse relationship between miRNA-29 and B7H3 protein level in normal or tumor tissues and cancer cell lines. (Xu H., et al., (2009), *Cancer Res.* 69:6275). In tumor patients, abnormal expression of B7H3 is closely associated with poor prognosis, increased tumor grade and metastasis, treatment resistance, and decreased overall survival rate of tumor patients (Picarda E., et al., (2016), *Clin. Cancer Res.* 22:3425).

[0006] The inhibitory effect of B7H3 on T cells, NK cells and DC cells significantly promotes immune evasion of tumor cells. In addition, B7H3 plays an important role in proliferation, migration, invasion, angiogenesis, epithelial to mesenchymal transition (EMT), cancer stemness and Warburg effect of tumor cells and tumor cell drug resistance. An increase in B7H3 can promote expression of Bcl-2 and Bcl-xl, to activate a JAK2-STAT3 signaling pathway, thereby enhancing anti-apoptotic activity of tumor cells (Zhang T., et al., (2015), *World J. Gastroenterol.* 21:1804). In addition, activation of AKT, ERK and JAK2/STAT3 pathways can induce expression of tumor cell metastasis-related protein factors, including MMP2, MMP9, CXCR4, and the like, thereby enhancing migration and aggressiveness of tumor cells (Tekle C., et al., (2013), *Int. J. Cancer* 130:2282; Li Y., et al., (2017), *Oncotarget* 8:71725; Wang L., et al., (2013), *PLOS One* 8: e70689; Liu F., et al., (2015), *Mol. Med. Rep.* 12:5455). Multiple studies have shown that B7H3 in tumor endothelial cells can also induce activation of NF- κ B through a TLR4-dependent mechanism, which significantly increases expression levels of VEGF and IL-8 (Tekle C., et al., (2012), *Int. J. Cancer* 130:2282), thereby promoting aggressiveness and angiogenesis of tumors (Ferrara N., et al., (2002), *Nat. Rev. Cancer* 2:795). In addition, B7H3 can also regulate activity of tumor cells EMT and cancer stem cells by reducing expression of E-cadherin and increasing expression of N-cadherin, vimentin, CD133 and CD44 (Jiang B., et al., (2016), *Oncotarget* 731755). Because B7H3 can activate STAT3 signaling to increase expression of hexokinase 2, B7H3 also has an effect of promoting

aerobic glycolysis in tumor cells (Shi T., et al., (2019), *Cell Death Dis.* 10:308). In addition, B7H3 inhibits activity of transcription factor NRF2, which increases levels of reactive oxygen species (ROS) and HIF1a, thereby further enhancing the aerobic glycolysis effect of tumor cells and promoting growth of tumor cells (Lim S., et al., (2016), *Cancer Res.* 76:2231). An increasing number of studies have shown that inhibiting or lowering expression of B7H3 can increase response of tumor cells to some drugs, for example, drugs for inhibiting DNA replication (including alkylating drugs) and drugs for inhibiting PI3K/Akt/mTOR and Ras/Raf/MEK signaling proteins (Flem-Karlsen K., et al., (2017), *Pigment Cell Melanoma Res.* 30:467; Kasten B. B., et al., (2017), *Nucl. Med. Biol.* 47:23; Liu H., et al., (2011), *Mol. Cancer Ther.* 10:960).

[0007] Transplanting tumor cells into mice lacking B7H3 or treating tumor mice with B7H3 antibodies significantly inhibits tumor growth (Cai D., et al., (2020), *Cell. Mol. Immunol.* 17:227; Lee Y. H., et al., (2017), *Cell Res.* 27:1034), indicating that blocking B7H3 signaling can be used for tumor treatment. In addition, using B7H3 antibodies in combination with anti-PD-1 antibodies can cause dual blockade of B7H3 and PD-1, thereby exerting a synergistic anti-tumor effect (Lee Y. H., et al., (2017), *Cell Res.* 27:1034). Due to a significant difference between B7H3 expression levels of normal tissues and tumor tissues, tumor cells can be effectively killed through the ADCC effect of B7H3 antibodies or toxin conjugation without causing excessive impact on normal tissues (Koenig S., et al., (2014), *Medicographia* 36:285). In summary, based on current research, B7H3 is a promising target for cancer treatment.

SUMMARY

[0008] In the present disclosure, a group of anti-B7H3 monoclonal antibodies were obtained through hybridoma technology. These antibodies have high B7H3 affinity, significant cellular internalization (allowing selective tumor killing or inhibition through toxin conjugation), significant antibody-dependent cytotoxicity (ADCC), or effective blocking of the immunosuppressive action of B7H3. In the present disclosure, six candidate antibodies were successfully humanized. These antibodies have extensive potential in developing drugs to inhibit cancer cells, regulate the action and levels of B7H3, and enhance immune response, especially in cancer treatment.

[0009] A murine or humanized B7H3 antibody or a functional fragment thereof provided in the present disclosure includes a heavy chain sequence and a light chain sequence. For anti-human B7H3 murine antibodies, the amino acid sequence information of the heavy chain variable regions and the light chain variable regions is as follows: The amino acid sequences of 7F5's heavy chain variable region and light chain variable region are SEQ ID NO: 1 and 2, respectively; the amino acid sequences of 9C8's heavy chain variable region and light chain variable region are SEQ ID NO: 3 and 4, respectively; the amino acid sequences of 5B6's heavy chain variable region and light chain variable region are SEQ ID NO: 5 and 6, respectively; the amino acid sequences of 7C9's heavy chain variable region and light chain variable region are SEQ ID NO: 7 and 8, respectively; the amino acid sequences of 2A9's heavy chain variable region and light chain variable region are SEQ ID NO: 9 and 10, respectively; the amino acid sequences of 4F11's heavy

chain variable region and light chain variable region are SEQ ID NO: 11 and 12, respectively; the amino acid sequences of 15A2's heavy chain variable region and light chain variable region are SEQ ID NO: 13 and 14, respectively; the amino acid sequences of 7B7's heavy chain variable region and light chain variable region are SEQ ID NO: 15 and 16, respectively; the amino acid sequences of 7E6's heavy chain variable region and light chain variable region are SEQ ID NO: 17 and 18, respectively; the amino acid sequences of 2E10's heavy chain variable region and light chain variable region are SEQ ID NO: 19 and 20, respectively; the amino acid sequences of 2F12's heavy chain variable region and light chain variable region are SEQ ID NO: 21 and 22, respectively; the amino acid sequences of 2F7's heavy chain variable region and light chain variable region are SEQ ID NO: 23 and 24, respectively; the amino acid sequences of 13A2's heavy chain variable region and light chain variable region are SEQ ID NO: 25 and 26, respectively; and the amino acid sequences of 14B3's heavy chain variable region and light chain variable region are SEQ ID NO: 27 and 28, respectively.

[0010] The amino acid sequence information of CDR1, CDR2, and CDR3 of the foregoing heavy chains and light chains of the antibodies is as follows: The amino acid sequences of 7F5's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 29, 30, and 31, respectively, and the amino acid sequences of 7F5's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 32, 33, and 34, respectively; the amino acid sequences of 9C8's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 35, 36, and 37, respectively, and the amino acid sequences of 9C8's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 38, 39, and 40, respectively; the amino acid sequences of 5B6's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 41, 42, and 43, respectively, and the amino acid sequences of 5B6's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 44, 45, and 46, respectively; the amino acid sequences of 7C9's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 47, 48, and 49, respectively, and the amino acid sequences of 7C9's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 50, 51, and 52, respectively; the amino acid sequences of 2A9's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 53, 54, and 55, respectively, and the amino acid sequences of 2A9's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 56, 57, and 58, respectively; the amino acid sequences of 4F11's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 59, 60, and 61, respectively, and the amino acid sequences of 4F11's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 62, 63, and 64, respectively; the amino acid sequences of 15A2's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 65, 66, and 67, respectively, and the amino acid sequences of 15A2's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 68, 69, and 70, respectively; the amino acid sequences of 7B7's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 71, 72, and 73, respectively, and the amino acid sequences of 7B7's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 74, 75, and 76, respectively; the amino acid sequences of 7E6's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 77, 78, and 79, respectively, and the amino acid sequences of 7E6's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 80, 81, and 82, respectively; the amino acid sequences of 2E10's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 83, 84, and 85, respectively, and the amino acid sequences of 2E10's light

chain CDR1, CDR2, and CDR3 are SEQ ID NO: 86, 87, and 88, respectively; the amino acid sequences of 2F12's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 89, 90, and 91, respectively, and the amino acid sequences of 2F12's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 92, 93, and 94, respectively; the amino acid sequences of 2F7's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 95, 96, and 97, respectively, and the amino acid sequences of 2F7's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 98, 99, and 100, respectively; the amino acid sequences of 13A2's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 101, 102, and 103, respectively, and the amino acid sequences of 13A2's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 104, 105, and 106, respectively; and the amino acid sequences of 14B3's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 107, 108, and 109, respectively, and the amino acid sequences of 14B3's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 110, 111, and 112, respectively.

[0011] Further, the anti-human B7H3 antibody or fragment is modified to become a humanized antibody.

[0012] An amino acid sequence of a heavy chain variable region of anti-human B7H3 humanized antibody 2A9 is SEQ ID NO: 113, and an amino acid sequence of a light chain variable region of anti-human B7H3 humanized antibody 2A9 is SEQ ID NO: 114; an amino acid sequence of a heavy chain variable region of humanized antibody 4F11 is SEQ ID NO: 115, and an amino acid sequence of a light chain variable region of humanized antibody 4F11 is SEQ ID NO: 116; an amino acid sequence of a heavy chain variable region of humanized antibody 9C8 is SEQ ID NO: 117, and an amino acid sequence of a light chain variable region of humanized antibody 9C8 is SEQ ID NO: 118; an amino acid sequence of a heavy chain variable region of humanized antibody 15A2 is SEQ ID NO: 119, and an amino acid sequence of a light chain variable region of humanized antibody 15A2 is SEQ ID NO: 120; an amino acid sequence of a heavy chain variable region of humanized antibody 5B6 is SEQ ID NO: 121, and an amino acid sequence of a light chain variable region of humanized antibody 5B6 is SEQ ID NO: 122; and an amino acid sequence of a heavy chain variable region of humanized antibody 7B7 is SEQ ID NO: 123, and an amino acid sequence of a light chain variable region of humanized antibody 7B7 is SEQ ID NO: 124.

[0013] An expression vector includes a nucleic acid molecule of the antibody.

[0014] A pharmaceutical composition includes the antibody or a functional fragment thereof, and a pharmaceutical carrier.

[0015] Use of the antibody or the functional fragment, nucleic acid molecule, expression vector, host cell, and pharmaceutical composition thereof in preparing B7H3 immunomodulatory medications is provided.

[0016] In the present disclosure, recombinant B7H3 receptor proteins produced in a mammalian cell expression system are used as antigens to immunize mice, followed by fusing the mice's splenocytes with myeloma cells to obtain hybridoma cells. Multiple monoclonal hybridoma cell lines are obtained through extensive cloning and screening of these hybridoma cells. These hybridoma cell lines can secrete monoclonal antibodies (FIG. 1 and FIG. 2) that specifically bind to B7H3 receptors. Some of these monoclonal antibodies have a significant effect of being endocy-

tosed by cells (FIG. 3 and FIG. 4), some have an effect of promoting human immune cells to secrete cytokines such as IFN- γ (FIG. 5), and some have a significant effect of mediating ADCC (FIG. 6, FIG. 7 and FIG. 8). Some monoclonal antibodies are potential candidates for future development of ADC, ADCC, and B7H3 antagonists. The invention further involves cloning the genes encoding antibody light and heavy chain variable regions through RT-PCR (Reverse Transcription-Polymerase Chain Reaction), and constructing humanized antibodies using the complementarity-determining regions graft (CDR-graft) method. In vitro functional experiments show that the humanized B7H3 antibodies can specifically bind to B7H3 receptor proteins (FIG. 9 and FIG. 10) while maintaining significant cellular internalization (FIG. 11), have a significant cytotoxic effect on Calu-6 tumor cells post-MMAE toxin conjugation (FIG. 12), an ADCC mediation effect (FIG. 13), a mediation effect on ADCC cytotoxic effect of NK92MI-hCD16 cells on Jurkat-B7H3 and PC9 cells (FIG. 14), and an effect of promoting human immune cells to produce and secrete cytokines such as IFN- γ (FIG. 15). The positive control antibodies used in the present disclosure are derived from patent applications U.S. Pat. No. 8,802,091 and WO2017180813A1.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] In order to illustrate the technical solutions in the specific embodiments of the present disclosure or in the prior art more clearly, the following briefly describes the accompanying drawings used for describing the specific embodiments or the prior art. Apparently, the accompanying drawings in the following description show only some embodiments of the present disclosure, and those of ordinary skills in the art may derive other drawings from these accompanying drawings without creative efforts.

[0018] FIG. 1(a-b): Determination of the binding properties of B7H3 hybridoma monoclonal antibodies with B7H3-hFc protein using ELISA method;

[0019] FIG. 2(a-b): Determination of the binding properties of B7H3 hybridoma monoclonal antibodies with PC9 cells using FACS method;

[0020] FIG. 3(a-b): Determination of the cellular internalization activity of B7H3 hybridoma antibodies by Jurkat-B7H3 cells through FACS;

[0021] FIG. 4(a-b): Determination of the cellular internalization activity of B7H3 hybridoma antibodies by PC9 cells through FACS;

[0022] FIG. 5(a-b): Promoting effect of B7H3 hybridoma antibodies on IFN- γ secretion in human PBMC cells;

[0023] FIG. 6(a-b): Testing of the ADCC effect on Jurkat-B7H3 cells mediated by B7H3 hybridoma or chimeric antibodies using the reporter gene method;

[0024] FIG. 7(a-b): Testing of the ADCC effect on PC9 tumor cells mediated by some B7H3 chimeric antibodies using the reporter gene method;

[0025] FIG. 8: Testing of the ADCC effect on DLD1 tumor cells mediated by some B7H3 chimeric antibodies using the reporter gene method;

[0026] FIG. 9: Determination of the binding properties of humanized B7H3 monoclonal antibodies with B7H3-hFc protein using ELISA method;

[0027] FIG. 10: Determination of the binding properties of humanized B7H3 monoclonal antibodies with B7H3 protein in PC9 cells using FACS method;

[0028] FIG. 11: Determination of the cellular internalization effect of humanized B7H3 monoclonal antibodies by PC9 cells through FACS;

[0029] FIG. 12: Cytotoxic effect of humanized B7H3 antibody ADC on Calu-6 tumor cells;

[0030] FIG. 13(a-b): Testing of the ADCC effect of humanized B7H3 antibodies on Jurkat-B7H3 and PC9 cells using the reporter gene method;

[0031] FIG. 14(a-b): ADCC cytotoxic effect of NK92MI-hCD16 cells on Jurkat-B7H3 and PC9 cells mediated by humanized B7H3 antibodies; and

[0032] FIG. 15: Promoting effect of humanized B7H3 monoclonal antibodies on IFN- γ production and secretion by human PBMC cells.

DETAILED DESCRIPTION

[0033] As used herein, the term “antibody” is used in its broadest sense and includes immunoglobulin or other molecules containing one or more antigen-binding domains that specifically bind to an antigen, namely, a protein or peptide that exhibit binding specificity for a particular antigen. Specific examples of antibodies may include intact antibodies (for example, classic four-chain antibody molecules), single-chain antibodies, single-domain antibodies, bispecific antibodies, multispecific antibodies, and the like. A classic antibody molecule is usually a tetramer composed of two identical heavy chains and two identical light chains connected to each other through disulfide bonds. Based on a conservative difference in amino acid sequences, heavy and light chains are divided into variable regions (V) located at amino terminals and constant regions (C) located at carboxyl terminals. The variable region is used to recognize and bind the antigen, and the constant region (for example, Fc fragment) is used to initiate a downstream effect, for example, cell-mediated antibody-dependent cytotoxicity (ADCC) effect. Within the variable regions of the heavy chain and light chain, three local regions have a higher degree of variation in amino acid composition and sequences, are key binding positions for antibodies and antigens, and therefore, are also referred to as complementarity-determining regions (CDRs). The amino acid sequences of the CDRs can be

readily determined by using numbering schemes recognized in the field, for example, Kabat, Chothia, IMGT, AbM or Contact.

[0034] An “antigen-binding fragment” of an antibody refers to an amino acid fragment in the antibody molecule that participates in specific antigen binding, for example, one of F(ab')₂, Fab and scFv.

[0035] EC₅₀ (concentration for 50% of maximal effect) refers to the concentration that causes 50% of the maximal effect. When used to indicate the binding ability of an antibody molecule to a corresponding antigen in an enzyme-linked immunosorbent assay (ELISA), EC₅₀ can refer to concentration of the antibody molecule when half of the maximum detection signal (for example, colorimetric or fluorescence intensity) occurs. The lower the value of EC₅₀, the greater the binding affinity to the antigen.

[0036] The present disclosure relates to an anti-B7H3 monoclonal antibody or an antigen-binding fragment thereof, including heavy chains and light chains, where an amino acid sequence of the heavy chain CDR1 is selected from one of SEQ ID NOs: 29, 35, 41, 47, 53, 59, 65, 71, 77, 83, 89, 95, 101 and 107; an amino acid sequence of heavy chain CDR2 is selected from one of SEQ ID NOs: 30, 36, 42, 48, 54, 60, 66, 72, 78, 84, 90, 96, 102 and 108; an amino acid sequence of heavy chain CDR3 is selected from one of SEQ ID NOs: 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91, 97, 103 and 109; an amino acid sequence of light chain CDR1 is selected from one of SEQ ID NOs: 32, 38, 44, 50, 56, 62, 68, 74, 80, 86, 92, 98, 104 and 110; an amino acid sequence of light chain CDR2 is selected from one of SEQ ID NOs: 33, 39, 45, 51, 57, 63, 69, 75, 81, 87, 93, 99, 105 and 111; and an amino acid sequence of light chain CDR3 is selected from one of SEQ ID NOs: 34, 40, 46, 52, 58, 64, 70, 76, 82, 88, 94, 100, 106 and 112, wherein the heavy and light chains of the antigen-binding fragment comprise amino acid sequences spanning from CDR1 to CDR3 of respective heavy and light chains of the antibody.

[0037] The anti-B7H3 antibody or antigen-binding fragment thereof provided in the present disclosure includes a heavy chain CDR and a light chain CDR, where the heavy chain CDR and the light chain CDR are selected from one or more groups of CDR combinations denoted as a to n:

Combination name	Heavy chain CDR1	Heavy chain CDR2	Heavy chain CDR3	Light chain CDR1	Light chain CDR2	Light chain CDR3
a	SEQ ID NO: 29	SEQ ID NO: 30	SEQ ID NO: 31	SEQ ID NO: 32	SEQ ID NO: 33	SEQ ID NO: 34
b	SEQ ID NO: 35	SEQ ID NO: 36	SEQ ID NO: 37	SEQ ID NO: 38	SEQ ID NO: 39	SEQ ID NO: 40
c	SEQ ID NO: 41	SEQ ID NO: 42	SEQ ID NO: 43	SEQ ID NO: 44	SEQ ID NO: 45	SEQ ID NO: 46
d	SEQ ID NO: 47	SEQ ID NO: 48	SEQ ID NO: 49	SEQ ID NO: 50	SEQ ID NO: 51	SEQ ID NO: 52
e	SEQ ID NO: 53	SEQ ID NO: 54	SEQ ID NO: 55	SEQ ID NO: 56	SEQ ID NO: 57	SEQ ID NO: 58
f	SEQ ID NO: 59	SEQ ID NO: 60	SEQ ID NO: 61	SEQ ID NO: 62	SEQ ID NO: 63	SEQ ID NO: 64
g	SEQ ID NO: 65	SEQ ID NO: 66	SEQ ID NO: 67	SEQ ID NO: 68	SEQ ID NO: 69	SEQ ID NO: 70
h	SEQ ID NO: 71	SEQ ID NO: 72	SEQ ID NO: 73	SEQ ID NO: 74	SEQ ID NO: 75	SEQ ID NO: 76
i	SEQ ID NO: 77	SEQ ID NO: 78	SEQ ID NO: 79	SEQ ID NO: 80	SEQ ID NO: 81	SEQ ID NO: 82
g	SEQ ID NO: 83	SEQ ID NO: 84	SEQ ID NO: 85	SEQ ID NO: 86	SEQ ID NO: 87	SEQ ID NO: 88

-continued

Combination name	Heavy chain CDR1	Heavy chain CDR2	Heavy chain CDR3	Light chain CDR1	Light chain CDR2	Light chain CDR3
k	SEQ ID NO: 89	SEQ ID NO: 90	SEQ ID NO: 91	SEQ ID NO: 92	SEQ ID NO: 93	SEQ ID NO: 94
l	SEQ ID NO: 95	SEQ ID NO: 96	SEQ ID NO: 97	SEQ ID NO: 98	SEQ ID NO: 99	SEQ ID NO: 100
m	SEQ ID NO: 101	SEQ ID NO: 102	SEQ ID NO: 103	SEQ ID NO: 104	SEQ ID NO: 105	SEQ ID NO: 106
n	SEQ ID NO: 107	SEQ ID NO: 108	SEQ ID NO: 109	SEQ ID NO: 110	SEQ ID NO: 111	SEQ ID NO: 112

[0038] In some embodiments, an amino acid sequence of a heavy chain variable region is selected from one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25 and 27, and an amino acid sequence of a light chain variable region is selected from one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26 and 28.

[0039] In some embodiments, the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof includes a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region and the light chain variable region are selected from one or more groups of variable region combinations denoted as A to N:

Combination name	Heavy chain variable region	Light chain variable region
A	SEQ ID NO: 1	SEQ ID NO: 2
B	SEQ ID NO: 3	SEQ ID NO: 4
C	SEQ ID NO: 5	SEQ ID NO: 6
D	SEQ ID NO: 7	SEQ ID NO: 8
E	SEQ ID NO: 9	SEQ ID NO: 10
F	SEQ ID NO: 11	SEQ ID NO: 12
G	SEQ ID NO: 13	SEQ ID NO: 14
H	SEQ ID NO: 15	SEQ ID NO: 16
I	SEQ ID NO: 17	SEQ ID NO: 18
J	SEQ ID NO: 19	SEQ ID NO: 20
K	SEQ ID NO: 21	SEQ ID NO: 22
L	SEQ ID NO: 23	SEQ ID NO: 24
M	SEQ ID NO: 25	SEQ ID NO: 26
N	SEQ ID NO: 27	SEQ ID NO: 28

[0040] Based on general understandings of persons skilled in the art, functional variants of the antibody or its antigen-binding fragment are also encompassed within the scope of this disclosure. The term “functional variant” used herein refers to a variant molecule obtained by introducing one or more amino acids for insertion, deletion or substitution based on the parent protein molecule (for example, a native protein molecule), where the variant molecule still retains at least some of the functions of the parent protein molecule (especially a function of interest, for example, an ability to bind to the corresponding antigen).

[0041] For the CDR region, the functional variants of the present disclosure include a heavy chain complementarity-determining region and a light chain complementarity-determining region. Compared with any one of the complementation-determining region groups denoted as a to n, the heavy chain complementarity-determining region and the light chain complementarity-determining region each contain mutations of up to 3 amino acids (for example, substi-

tution, deletion or addition of 1, 2 or 3 amino acids or any combination thereof). Preferably, the mutations are conservative mutations.

[0042] For the variable region, in some embodiments, the functional variant in the present disclosure includes a heavy chain variable region, and an amino acid sequence of the heavy chain variable region includes a sequence sharing at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identity with any one of the heavy chain variable region sequences shown in the groups A to N. In some embodiments, the functional variant in the present disclosure includes a light chain variable region, and an amino acid sequence of the light chain variable region includes a sequence sharing at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identity with any one of the light chain variable region sequences shown in the groups A to N. The term “identity” refers to a quantity of a degree of identity between two amino acid or nucleotide sequences (for example, a query sequence and a reference sequence), and is generally expressed as a percentage. Usually, before the percentage of identity between two amino acid or nucleotide sequences is calculated, the two amino acid or nucleotide sequences are first subjected to sequence alignment and a gap (if any) is introduced. At a certain alignment position, if amino acid residues or bases in the two sequences are the same, the two sequences are consistent or matched at the position; or if the amino acid residues or bases in the two sequences are different, the two sequences are considered to be inconsistent or mismatched at the position. In some algorithms, the number of matched positions is divided by the total number of positions in the alignment window to obtain the sequence identity. In other algorithms, the number of gaps and/or gap length is also taken into account. Commonly used sequence alignment algorithms or software include DANMAN, CLUSTALW, MAFFT, BLAST, MUSCLE, among others. For the purposes of the present disclosure, the publicly available alignment software BLAST (available at <https://www.ncbi.nlm.nih.gov/>) can be used to obtain optimal sequence alignment through default settings and calculate the sequence identity between two amino acid or nucleotide sequences.

[0043] The functional variant retains the ability to specifically bind B7H3. Persons skilled in the art can determine a suitable variant of the antigen-binding molecule illustrated herein by using well-known technologies. In certain embodiments, persons skilled in the art can identify regions in the antibody or antigen-binding fragment thereof that are unimportant for the activity of specifically binding B7H3 and change suitable regions without destroying the activity.

[0044] In some embodiments, the B7H3 monoclonal antibody or antigen-binding fragment thereof is murine or humanized.

[0045] In some embodiments, the heavy chain variable region and light chain variable region are humanized.

[0046] In some embodiments, the heavy and light chains are humanized.

[0047] The term “humanized antibody”, also known as CDR-grafted antibody, refers to an antibody produced by transplanting the first animal-derived CDR sequence into human antibody variable region frameworks, that is, different types of human germline antibody framework sequences. Humanized antibodies can overcome the heterologous reactions induced by chimeric antibodies due to their significant murine protein content. Such framework sequences can be obtained from public DNA databases or published references containing germline antibody gene sequences. For example, germline DNA sequences of human heavy and light chain variable region genes are available in “VBase” human germline sequence database (www.mrcpe.com.ac.uk/vbase) and found in Sequences of Proteins of Immunological Interest (Kabat, E. A. et al., 1991, 5th ed.) To avoid both a decrease in immunogenicity and a resulting decrease in activity, a minimum of reverse mutation or back mutation can be performed on the human antibody variable region framework sequence to maintain activity. The humanized antibody in the present disclosure also includes a mature humanized antibody obtained after phage shows affinity to the CDR. In a preferred embodiment of the present disclosure, the first animal origin is of murine origin.

[0048] In some embodiments, the heavy chain and the light chain are humanized; an amino acid sequence of a humanized heavy chain variable region is selected from one of SEQ ID NOs: 113, 115, 117, 119, 121, and 123; and the amino acid sequence of the humanized light chain variable region is selected from one of SEQ ID NOs: 114, 116, 118, 120, 122, and 124.

[0049] In some embodiments, the humanized heavy chain variable region and light chain variable region are selected from one or more groups of variable region combinations denoted as 1 to 6:

Combination name	Heavy chain variable region	Light chain variable region
1	SEQ ID NO: 113	SEQ ID NO: 114
2	SEQ ID NO: 115	SEQ ID NO: 116
3	SEQ ID NO: 117	SEQ ID NO: 118
4	SEQ ID NO: 119	SEQ ID NO: 120
5	SEQ ID NO: 121	SEQ ID NO: 122
6	SEQ ID NO: 123	SEQ ID NO: 124

[0050] In some embodiments, the heavy chain and the light chain are humanized; an amino acid sequence of a humanized heavy chain variable region is an amino acid sequence with SEQ ID NO: 113; and the amino acid sequence of the humanized light chain variable region is an amino acid sequence with SEQ ID NO: 114.

[0051] In some embodiments, the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof includes a constant region of human IgG1.

[0052] The present disclosure further relates to use of the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof in preparation of the following medication: a medication for regulating B7H3 activity or B7H3 level, a

medication that has significant cellular internalization (allowing selective tumor killing or inhibition through toxin conjugation), significant antibody-dependent cytotoxicity (ADCC), or an ability to block B7H3 in immunosuppression to enhance body immunity, a medication for promoting T lymphocyte, or a medication for improving production of cytokine such as IFN- γ in T lymphocyte.

[0053] The present disclosure further relates to a monoclonal antibody conjugate, including a monoclonal antibody and a conjugating part, wherein the monoclonal antibody is one of the previously mentioned anti-B7H3 monoclonal antibodies or their antigen-binding fragments, and the conjugating part is selected from one or more of a radionuclide, a medication, a toxin, a cytokine, a cytokine receptor fragment, an enzyme, fluorescein, and biotin.

[0054] The present disclosure further relates to use of the monoclonal antibody conjugate in preparation of the following medication: a medication for regulating B7H3 activity or B7H3 level, a medication having significant cellular internalization (allowing selective tumor killing or inhibition through toxin conjugation), a medication having significant antibody-dependent cytotoxicity (ADCC), a medication for blocking B7H3 in immunosuppression to enhance body immunity, a medication for promoting T lymphocyte, or a medication for improving production of cytokine such as IFN- γ in T lymphocyte.

[0055] The present disclosure further relates to use of the foregoing monoclonal antibody conjugate in preparation of a medication for preventing, treating, and/or adjuvantly treating a tumor.

[0056] The present disclosure further relates to an expression vector, including a nucleic acid molecule of the antibody.

[0057] The present disclosure further relates to an expression vector, including a nucleic acid molecule of the foregoing anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof.

[0058] The term “vector” refers to a nucleic acid delivery vehicle into which a polynucleotide can be inserted. When a vector enables an inserted polynucleotide-encoded protein to be expressed, the vector is referred to as an expression vector. The vector can enter a host cell via transformation, transduction, or transfection, so that an element of a genetic material carried by the vector is expressed in the host cell. The vector is well-known to a person skilled in the art and includes, but is not limited to, a plasmid, a phagemid, a cosmid, an artificial chromosome (for example, a yeast artificial chromosome (YAC), a bacterial artificial chromosome (BAC), or a P1 derived artificial chromosome (PAC)), a phage (λ phage or M13 phage), and an animal virus. An animal virus that can serve as the vector includes, but is not limited to, retrovirus (including lentivirus), adenovirus, adeno-associated virus, herpesvirus (for example, herpes simplex virus), poxvirus, baculovirus, papillomavirus, and papillomavirus (for example, SV40). In some embodiments, the vector in the present disclosure includes a regulatory element commonly used in genetic engineering, for example, the enhancer, the promoter, an internal ribosome entry site (IRES) and another expression control element (for example, a transcription termination signal, or a polyadenylation signal and a polyU sequence).

[0059] In the present disclosure, the vector can be a composition, for example, a mixture of multiple plasmids, where different plasmids carry part of the antibody or the antigen-binding fragment.

[0060] The present disclosure further relates to a pharmaceutical composition, including the foregoing antibody or a (at least one) functional fragment thereof, and a pharmaceutical carrier.

[0061] As used herein, the “pharmaceutical carrier” includes any material that allows a component to remain biologically active when combined with the active component and that does not react with an immune system of the subject. An example includes, but is not limited to, any one of a standard drug carrier (for example, a phosphate buffered saline solution, water, an emulsion (for example, an oil/water emulsion)) and various types of wetting agents. A composition containing such carrier is prepared in a well-known conventional method (refer to, for example, Remington’s Pharmaceutical Sciences, 18th edition, A. Gennaro, ed., Mack Publishing Co., Easton, PA, 1990; and Remington, The Science and Practice of Pharmacy, 21st edition, Mack Publishing, 2005).

[0062] The present disclosure further relates to use of the foregoing antibody or the functional fragment, nucleic acid molecule, expression vector, host cell, and pharmaceutical composition thereof in preparing B7H3 immunomodulatory medications.

[0063] The embodiments of the present disclosure are described in detail below with reference to examples:

Example 1

Immunization and Cell Fusion in Mice for Producing Anti-B7H3 Antibodies

[0064] Human B7H3 (NCBI Reference Sequence: NM_001024736.2) extracellular domain and mFc fusion protein (B7H3-ECD-mFc) were used as antigens, and fully emulsified with an equal volume of complete Freund’s adjuvant (Sigma, Cat. No.: F5581), 6-8 week old Balb/c mice (Zhaoyan (Suzhou) New Drug Research Center Co., Ltd.) were subcutaneously immunized with a dose of 50 µg antigen per mouse. Subsequently, the mice were immunized subcutaneously three times with the same dose of antigen fully emulsified in incomplete Freund’s adjuvant (Sigma, Cat. No.: F5506) every 2 weeks. After three immunizations, serum titers of the mice were measured, and the mice were booster immunized once via intraperitoneal injection 3 days before the fusion. Using PEG Hybri-Max (Sigma, Cat.: 7181) as a fusion agent, splenocytes of the mice and SP2/0 cells were mixed in a ratio of 4:1. Fused cells were added into a 96-well plate (1×10^5 cells/well), and each well contained 0.1 mL of 1×HAT medium (Invitrogen, Cat. No.: 21060-017). 0.1 mL of HT (Invitrogen, Cat. No.: 11067-030) medium was added on day 3, the medium was pipetted out of the 96-well plate on day 7, and 0.2 mL of fresh HT medium was added. The supernatant was collected on day 9 for various screening and tests.

Example 2

Antigen Binding Ability of Hybridoma Antibodies and Subcloning

[0065] The binding ability of hybridoma antibodies was tested using ELISA and FACS, along with subcloning via the limiting dilution method:

(1) Screening of B7H3-binding positive clones through ELISA: 50 µL of B7H3-hFc (final concentration: 2 µg/mL) was used to coat each well of a 96-well ELISA plate (Corning, Cat. No.: 9018), and incubated overnight at room temperature. After the ELISA plate was rinsed with washing buffer (PBS+0.05% Tween20) 3 times, blocking buffer (PBS+2% BSA (Sigma, Cat. No.: V90093)) was added and the ELISA plate was incubated at room temperature for 1 hour. The ELISA plate was rinsed with washing buffer 3 times; and hybridoma supernatant was added, and the ELISA plate was incubated at room temperature for 1 hour and rinsed 3 times. 100 µL of 10,000-fold diluted HRP-conjugated goat anti-mouse IgG secondary antibody (Thermo, Cat. No.: 31432) was added to each well, and the ELISA plate was incubated at room temperature for 1 hour in the dark and rinsed 3 times. 100 µL of TMB (Biocytogen, Cat. No.: ES-002) was added to each well, and the ELISA plate was incubated at room temperature for color development for 2 minutes, 100 µL of reaction terminating solution (2N H₂SO₄) was added to each well to terminate the color reaction, and an OD450 value of each well was read using a microplate reader (Tecan Spark).

(2) Screening of B7H3-binding positive clones through FACS: 50 µL of the positive hybridoma supernatant or purified hybridoma antibody in the foregoing test was mixed with 50 µL of 293T-B7H3 cells (2×10^5 cells/well), and a resulting mixture was added to a 96-well plate with a U-shaped bottom and left standing at 4° C. for 1 hour, the 96-well plate was rinsed with FACS buffer (PBS+3% FCS) and centrifuged twice, then 400-fold diluted PE-labeled goat anti-mouse secondary antibody (Biolegend, Cat. No.: 405307) was added, the 96-well plate was incubated at 4° C. for 30 minutes in the dark, rinsed with FACS buffer and centrifuged twice, and signal values of the cells in a PE channel were detected by using BD Accuri C6 flow cytometer.

(3) Subcloning: Positive multiclonal hybridoma cells identified by ELISA and FACS were subcloned using the limiting dilution method. The subclones were further detected and screened using ELISA and FACS to obtain positive hybridoma monoclonal clones.

Antibody Purification, Concentration Determination, and Endotoxin Testing:

[0066] The positive monoclonal hybridoma cells were added to 50 ml of serum-free medium (Invitrogen, Cat. No.: 12045-076) and the serum-free medium was cultured for 8 to 9 days, and then centrifuged to collect supernatant. The monoclonal antibodies were purified through protein A affinity chromatography. The purified antibody samples were liquid-exchanged and concentrated in ultrafiltration centrifuge tubes (Millipore, Cat. No.: ACS500024). Protein concentration was determined in a BCA method and limulus reagent (Xiamen Bioendo Technology Co., Ltd.) was used to detect endotoxin content of purified antibody samples.

[0067] ELISA and FACS were conducted to detect the binding ability of purified antibody samples to B7H3. Results are shown in FIG. 1(a-b), FIG. 2(a-b) and Table 1. The selected hybridoma antibodies had higher affinity.

TABLE 1

EC ₅₀ values for the binding of B7H3 hybridoma monoclonal antibodies to the B7H3 antigen														
Parameter	Antibody													
	7F5	9C8	5B6	7C9	2A9	4F11	15A2	7B7	7E6	2E10	2F12	2F7	13A2	14B3
ELISA EC ₅₀ (ng/mL)	5.76	10.45	4.93	9.9	6.24	5.11	1.51	9.7	4.49	6.7	na	2.78	0.52	9.1
FACS EC ₅₀ (ng/mL)	197	86.1	71.3	103.5	186.5	83.4	6.2	179.2	52.6	234	217	177	100.7	248.5

Example 3

Testing Intracellular Phagocytosis and B7H3 Blocking Action of Hybridoma Monoclonal Antibodies

[0068] Analysis of intracellular phagocytosis of antibodies: In a 96-well plate, 50 μ L of 2×10^6 /mL PC9 cells and 50 μ L of labeled B7H3 antibodies were added to each well, and the 96-well cell culture plate was incubated at 4° C. for 1 hour. After the cell culture plate was rinsed with FACS buffer twice, Goat anti-mFc (Jackson, Cat. No.: 115-005-071) secondary antibody labeled with pH-dependent fluorescent dye CypHer5E (GE, Cat. No.: PA15401) was added, and the cell culture plate was incubated at 4° C. for 0.5 hours and rinsed twice. A culture medium (1640+10% FBS) was added, the cell culture plate was put in an incubator and incubated for 3 hours, and then centrifuged to remove the culture medium. A residue was resuspended in PBS at pH 9.0, and CypHer5E signal of PC9 cells was detected through a BD C6 flow cytometer. Efficiency of entering the cells by B7H3 antibodies was calculated. As shown in FIG. 3(a-d) and FIG. 4(a-d), Jurkat-B7H3 and PC9 cells had significant endocytic activity for B7H3 antibodies, including 15A2, 5B6, 7C9, 2F7, 7B7, 2E10 and 4F11.

[0069] Testing of the effect of B7H3 hybridoma antibodies on secretion of cytokines by human PBMC cells: PBMC cells (TPCS, Cat. No.: PB025C) resuspended in complete medium (RPMI1640+10% FCS) was added into the 96-well plate (Corning, Cat. No.: 3799), then 40 ng/ml OKT3 (eBioscience, Cat. No.: 16-0037-85) was added, and the 96-well plate was incubated at 37° C. for 72 hours. After counting, the activated PBMC cells were resuspended in complete culture medium (2.5×10^5 cells/mL). In the 96-well plate, 100 μ L of PBMC cells and 50 μ L of B7H3 antibodies at different concentrations (initial concentration of 20

μ g/mL, 10-fold serial dilution) were added to each well, the 96-well cell culture plate (Corning, Cat. No.: 3599) was placed in an incubator containing 5% CO₂ at 37° C. for 48 hours, and the supernatant was collected. The concentrations of cytokines were detected using IFN- γ ELISA kit (R&D Systems, Cat. No.: DY285). As shown in FIG. 5, B7H3 antibodies including 5B6, 2F12, 2F7, 15A2, 13A2, 14B3, 4F11, 2A9, etc., can significantly promote the secretion of IFN- γ by PBMC cells.

Example 4

Cloning and Sequencing of B7H3 Antibody Variable Region Genes

[0070] B7H3 monoclonal hybridoma cell lines were lysed with TRIZON (Cwbiotech, Cat No: CW0580), and the total RNAs of hybridoma cells were extracted. RNA of hybridoma cells was reversely transcribed into cDNA by using HiFi Script cDNA Synthesis Kit (Cwbiotech, Cat No: CW2569). The cDNA was used as a template, degenerate primers were used to amplify heavy chain and light chain variable region genes of the antibody in a PCR method (Kettleborough et al. (1993) Eur. J Immunology 23:206-211; Strebe et al. (2010) Antibody Engineering 1:3-14). After the product of the PCR amplification was ligated to T/A vector, DH5 α competent cells were transformed, inoculated and cultured overnight at 37° C. Monoclonal was picked from the culture plate and cultured for amplification, plasmids were extracted, and the gene sequence of the antibody was identified. Based on the gene sequence of the antibody, complementarity-determining regions (CDRs) and framework regions of the antibody were analyzed. The B7H3 antibody sequence number is shown in Table 2, and specific sequence information is shown in the sequence listing.

TABLE 2

B7H3 Antibody Sequence Number			
SEQ ID	Description		
NO.	Antibody	Sequence description	Amino acid sequence
1	7F5	Amino acid sequence of a heavy chain variable region	EVQLQQSGAELVRPGALVKLSCKTSGF NIKAYYIHWVKQRPEQGLEWIGWIDPE NGNTFYDPKFGQKASITADTSSNTAYLQ LSSLTSEDAVYYCTRSSTRLRPFQYD GQGTTLTVSS
2	7F5	Amino acid sequence of a light chain variable region	DIVMTQAAFSNPVTLTGTSASISCRSSKS LLHSNGITYLYWYLQKSGQSPQLLIYQM SNLASGVDPDRFSSSGSGTDFTLRISR V EAEDVGVIYCAQNLELYTFGGGKLEI K

TABLE 2-continued

B7H3 Antibody Sequence Number			
SEQ ID	Description		
NO.	Antibody	Sequence description	Amino acid sequence
3	9C8	Amino acid sequence of a heavy chain variable region	EVQLQQSGPELVKPGASVKISCKTSGY TFTEYTIHWVKQSHGKSLEWIGGINPNS GGTTYNLKFKGKATLTVDKSSSTVYME LRSLTSEDSAVYYCARKEGFFTTDYWG QGTTLTVSS
4	9C8	Amino acid sequence of a light chain variable region	QTVLTQSPAIMASAPGEKVTMTCSASS SLSYIHWYRQKSGTSPKRWIYDTSNLA TGVFVRFSGSGSGTSYSLTISSEAEAD AATYYCQQWHYNPPTFGGGTKLEIK
5	5B6	Amino acid sequence of a heavy chain variable region	QVQLQQSGAELARPGASVKMSCKTSG YTFITYTIHWVKQRPQGLEWIGYINPT SDYANFNQKPKDKATLTADKSSRTAYM QLSSLTSEDSAVYYCARWGTTFVQFYF DYWGQGTTLTVSS
6	5B6	Amino acid sequence of a light chain variable region	QIVLSQSPAILASAPGEKVTMTCRASQS LSYINWYQLKPGSSPKPWYATSSLQS GVPARFSGSASGTSYSLTISRVEAEDAA TYFCQHWSSDPPTIGGGTKLEIK
7	7C9	Amino acid sequence of a heavy chain variable region	EVQLQQSGTELVRPGALVKLSCKASAL NIKDFYMHVVKQRPERGLEWIGWIDPD NGNTIYDPKFGKASITADISSSTAYLQL SSLTSEDYAVYYCARSSSTRQRPFDYWG QGTTLTVSS
8	7C9	Amino acid sequence of a light chain variable region	DIVMTQSHKVMSTSVGDRVSITCKASQ DVRTGVAWYQKQKPGQSPKLLIYSAFYR YTGVPDRFTVSGSGADFTFTITSVQAE DLAVYYCQQYYSIPWTFGGGKLEIK
9	2A9	Amino acid sequence of a heavy chain variable region	EVQLQQSGPELVKPGASVKISCKTSGY TFTEYIMHWVKQSRGKSLEWIGGINPN SGGTTYNEKFKGKATLTVDKSASTAYM ELRSLTSEDSAIYYCARRMPYWFYFDVW GAGTTVTVSS
10	2A9	Amino acid sequence of a light chain variable region	QIVLTQSPAIMASAPGEKVTMTCSASS VSYIHWYRQKSGTSPKRWVYDTSKLAS GVPARFSGSGSGTSYSLTISSEAEADA ATYYCQQWVSNPLTFGAGTKLELK
11	4F11	Amino acid sequence of a heavy chain variable region	EVQLQQSGPELVKPGASVKLSCKTSGY TFTEYIMHWVKQSHGKSLEWIGGINPN TGGITYNQKFKDKATLTVDKSSSTAYME LRSLTSDSAVYFSARAGGNFLWYFDV WGAGTTVTVSS
12	4F11	Amino acid sequence of a light chain variable region	QIVLTQSPATMASAPGEKVTMTCSASS SVSYMHWYQKSDMSPKRWLYDTSKL ASGVPARFSGSGSGTSYSLAISSEAE DAATYYCQQWNSNPLTFGTGKLELK
13	15A2	Amino acid sequence of a heavy chain variable region	QVQLQQSGPEVVKPGALVRISCKASGY TFTSYDINWVKQRPQGLEWIGWIDPG DGSTKYNEKFKGKATLTADKSSSTAYM QFSSLTSENSAVYFCVRRGFDYWGQG TTLTVSS
14	15A2	Amino acid sequence of a light chain variable region	QILLTQSPAIMASAPGEKVTMTCRTSSS VSYLHWYQKPGSSPKLWIHDTSNLAS GFPARFSGSGSGTSYSLIISSEAEADA TYCHQRSSYPWTFGGGKLEIK

TABLE 2-continued

B7H3 Antibody Sequence Number			
SEQ ID	Description		
NO.	Antibody	Sequence description	Amino acid sequence
15	7B7	Amino acid sequence of a heavy chain variable region	EVKLVESGGGLVRPGGSLKLSCAASGF TFSSYAVSWVRQTPEKRLEWVASISGG GIYIYYPDSVKGRFTISRDNKNNLYLQ MNSLTPDDTALYYCARHGGAGYFDYW GQGTTLTVSS
16	7B7	Amino acid sequence of a light chain variable region	DSLMTQSPASLSASVGETVTITCRGSE SVHSYLAHYQQKQKSPQLLVYNAKTL QSGVPSRFSGGSGTGQFSLRINSLQPE DFGTYYCQQHYGSPFWTFGGGTILEIK
17	7E6	Amino acid sequence of a heavy chain variable region	QVQLQQSGPELVRPGVSVKISCKGSGY TFTDYAMHWVKQSHAKSLEWIGVINSY YGNTHYNQKFKGKATMTVDKSSSTAYM ELARLTSEDSAIYYCARAGGYSYGSSY RPDRFAYWGQGLTVISA
18	7E6	Amino acid sequence of a light chain variable region	DIQMTQSPASQSASLGESVTITCLASQT IGTWLAHYQQKPKGKSPQLLIYAATNLAD GVPSRFSGGSGTGKFSFKISLQAEDE VSYYCQQLYSTPYTFGGGTLEIK
19	2E10	Amino acid sequence of a heavy chain variable region	EVQLQQSGPELVKPGASVKISCKTSGY TFTEYTMHWVKQSHGKNLEWIGGINTD NGGTTYQKFKDKATLTVDRSSSTAYM VLRSLTSEDSAVYYCARSFNNYGPWF AYWGQALVTISA
20	2E10	Amino acid sequence of a light chain variable region	DIVMTQSHKFMSTSIGDRVSITCKASQD VRSVAWAYQRPQGSPKLLIYWASTRH TGVPARFTGSGSGTDYTLTINNVQSED LAGYFCQQYSSYFTFGGGTKLEIK
21	2F12	Amino acid sequence of a heavy chain variable region	EVQLQQSGPELVKSGASVNISCKTSGY TFTENTMHWVKLSHGKSLEWIGGINPN NGGTTYQKFKGKATLTVDKSSSTAYM ELRSLTSEDSAVYYCARRGRWPPDWY FDVWGAGTTVTSS
22	2F12	Amino acid sequence of a light chain variable region	QIVLTQSPAIMSASPGEKVTMTCSASS LSYMHVYQQKSGTSPKRWIYDTSKLA SGVPARFSGSGSTSYSLTISMEAEDE AATYYCQQWSNPLTFGAGTKLEIK
23	2F7	Amino acid sequence of a heavy chain variable region	QVQLQQGAELVKPGASVKLSCKASG YTFTSYIYWLKQRPQGLEWIGGIKPS NGDTNFMNAKFMNKATLTVDKSSSTAYM QLSSLTSEDSAIYYCTRFYGYSPYDFDV WGAGTTVTSS
24	2F7	Amino acid sequence of a light chain variable region	NIVMTQSPKSMMAVGERVTLTKASE NVDTHVSWYQQKPEQSPTLLMYGTSN RFTGVPRFTGSGSATDFTLTISVQAE DLADYHCGQSYRYPYTFGGGTLEIK
25	13A2	Amino acid sequence of a heavy chain variable region	EVQLQQSGPELVKPGASVKISCKTSGY TFTEYVMHWVKQRHGQSLWIGGINP NTGGTTYQKFKGKATLTVDKSSSSAY MEFRSLTSEDSAVYYCARRMPFGISAM DYWGQTSVTSS
26	13A2	Amino acid sequence of a light chain variable region	QIVLTQSPAIMSASPGEKVTMTCSASS VSYIHWYQQKSGTSPKRWISETSKLAS GVPARFSGSGSTSYSLTISMGAEDE ATYYCHQWISYPPTFGAGTKLEIK

TABLE 2-continued

B7H3 Antibody Sequence Number			
SEQ ID	Description		
NO.	Antibody	Sequence description	Amino acid sequence
27	14B3	Amino acid sequence of a heavy chain variable region	QVQLQQSGPELVKSVKISCKGSGY TFTDHAHWWKQSHAKSLEWIGVINTYS GNTNFNQKLKGKATMTVDKSSSTAYME LARLTSEDSAIYYCARGAHSSGYPYWY FDVWGAGTTTVTVSS
28	14B3	Amino acid sequence of a light chain variable region	QIVLSQSPAILASASAGEKVTMTCRASSS VSYMHWYQQKPGSSPKPWIYATSNLA SGVPARFSGSGSGTSYSLTISRVEAED AATYYCQQWSTNPFTFGSGTKLEMK
29	7F5	Amino acid sequence of a heavy chain CDR1 region	AYYIH
30	7F5	Amino acid sequence of a heavy chain CDR2 region	WIDPENGNTFYDPKFQG
31	7F5	Amino acid sequence of a heavy chain CDR3 region	SSTRLRPFDY
32	7F5	Amino acid sequence of a light chain CDR1 region	RSSKSLHLSNGITYLY
33	7F5	Amino acid sequence of a light chain CDR2 region	QMSNLAS
34	7F5	Amino acid sequence of a light chain CDR3 region	AQNLELYT
35	9C8	Amino acid sequence of a heavy chain CDR1 region	EYTIH
36	9C8	Amino acid sequence of a heavy chain CDR2 region	GINPNSGGTTYNLKFKG
37	9C8	Amino acid sequence of a heavy chain CDR3 region	KEGFFTDY
38	9C8	Amino acid sequence of a light chain CDR1 region	SASSLSYIH
39	9C8	Amino acid sequence of a light chain CDR2 region	DTSNLAT
40	9C8	Amino acid sequence of a light chain CDR3 region	QQWHYNPPT
41	5B6	Amino acid sequence of a heavy chain CDR1 region	TYTIH
42	5B6	Amino acid sequence of a heavy chain CDR2 region	YINPTSDYANFNQKFKD
43	5B6	Amino acid sequence of a heavy chain CDR3 region	WGTTVFQFYFDY
44	5B6	Amino acid sequence of a light chain CDR1 region	RASQSLSYIN
45	5B6	Amino acid sequence of a light chain CDR2 region	ATSSLQS
46	5B6	Amino acid sequence of a light chain CDR3 region	QHWSSDPPT
47	7C9	Amino acid sequence of a heavy chain CDR1 region	DFYMH
48	7C9	Amino acid sequence of a heavy chain CDR2 region	WIDPDNGNTIYDPKFQG

TABLE 2-continued

B7H3 Antibody Sequence Number			
SEQ ID	Description		
NO.	Antibody	Sequence description	Amino acid sequence
49	7C9	Amino acid sequence of a heavy chain CDR3 region	SSTRQRPFDY
50	7C9	Amino acid sequence of a light chain CDR1 region	KASQDVRTGVA
51	7C9	Amino acid sequence of a light chain CDR2 region	SAFYRYT
52	7C9	Amino acid sequence of a light chain CDR3 region	QQYYSIPWT
53	2A9	Amino acid sequence of a heavy chain CDR1 region	EYIMH
54	2A9	Amino acid sequence of a heavy chain CDR2 region	GINPNSGGTTYNEKPKG
55	2A9	Amino acid sequence of a heavy chain CDR3 region	RMPYWYFDV
56	2A9	Amino acid sequence of a light chain CDR1 region	SASSSVSYIH
57	2A9	Amino acid sequence of a light chain CDR2 region	DTSKLAS
58	2A9	Amino acid sequence of a light chain CDR3 region	QQWVSNPLT
59	4F11	Amino acid sequence of a heavy chain CDR1 region	EYIMH
60	4F11	Amino acid sequence of a heavy chain CDR2 region	GINPNTGGITYNQKPKD
61	4F11	Amino acid sequence of a heavy chain CDR3 region	AGGNFLWYFDV
62	4F11	Amino acid sequence of a light chain CDR1 region	SASSSVSYM
63	4F11	Amino acid sequence of a light chain CDR2 region	DTSKLAS
64	4F11	Amino acid sequence of a light chain CDR3 region	QQWNSNPLT
65	15A2	Amino acid sequence of a heavy chain CDR1 region	SYDIN
66	15A2	Amino acid sequence of a heavy chain CDR2 region	WIDPGDGSTKYNEKPKG
67	15A2	Amino acid sequence of a heavy chain CDR3 region	RGFDY
68	15A2	Amino acid sequence of a light chain CDR1 region	RTSSSVSYLH
69	15A2	Amino acid sequence of a light chain CDR2 region	DTSNLAS
70	15A2	Amino acid sequence of a light chain CDR3 region	HQRSSYPWT
71	7B7	Amino acid sequence of a heavy chain CDR1 region	SYAVS

TABLE 2-continued

B7H3 Antibody Sequence Number			
SEQ ID	Description		
NO.	Antibody	Sequence description	Amino acid sequence
72	7B7	Amino acid sequence of a heavy chain CDR2 region	SISGGGIYIYPDSVKG
73	7B7	Amino acid sequence of a heavy chain CDR3 region	HGGAGYFDY
74	7B7	Amino acid sequence of a light chain CDR1 region	RGSESVHSYLA
75	7B7	Amino acid sequence of a light chain CDR2 region	NAKTLQS
76	7B7	Amino acid sequence of a light chain CDR3 region	QQHYGSPPW
77	7E6	Amino acid sequence of a heavy chain CDR1 region	DYAMH
78	7E6	Amino acid sequence of a heavy chain CDR2 region	VINSYYGNTHYNQKFKG
79	7E6	Amino acid sequence of a heavy chain CDR3 region	AGGYSYGSSYRPFAY
80	7E6	Amino acid sequence of a light chain CDR1 region	LASQTIGTWLA
81	7E6	Amino acid sequence of a light chain CDR2 region	AATNLAD
82	7E6	Amino acid sequence of a light chain CDR3 region	QQLYSTPYT
83	2E10	Amino acid sequence of a heavy chain CDR1 region	EYTMH
84	2E10	Amino acid sequence of a heavy chain CDR2 region	GINTDNGGTTYSQKFKD
85	2E10	Amino acid sequence of a heavy chain CDR3 region	SFNNYGPWFAY
86	2E10	Amino acid sequence of a light chain CDR1 region	KASQDVRSVA
87	2E10	Amino acid sequence of a light chain CDR2 region	WASTRHT
88	2E10	Amino acid sequence of a light chain CDR3 region	QQYSSYFT
89	2F12	Amino acid sequence of a heavy chain CDR1 region	ENTMH
90	2F12	Amino acid sequence of a heavy chain CDR2 region	GINPNNGGTTYNQKFKG
91	2F12	Amino acid sequence of a heavy chain CDR3 region	RGRWPPDWYFDV
92	2F12	Amino acid sequence of a light chain CDR1 region	SASSLSYMH
93	2F12	Amino acid sequence of a light chain CDR2 region	DTSKLAS
94	2F12	Amino acid sequence of a light chain CDR3 region	QQWSSNPLT

TABLE 2-continued

B7H3 Antibody Sequence Number			
SEQ ID	Description		
NO.	Antibody	Sequence description	Amino acid sequence
95	2F7	Amino acid sequence of a heavy chain CDR1 region	SYIY
96	2F7	Amino acid sequence of a heavy chain CDR2 region	GIKPSNGDTNFNKFMN
97	2F7	Amino acid sequence of a heavy chain CDR3 region	FFYGSPYDFDV
98	2F7	Amino acid sequence of a light chain CDR1 region	KASENVDTHVS
99	2F7	Amino acid sequence of a light chain CDR2 region	GTSNRFT
100	2F7	Amino acid sequence of a light chain CDR3 region	GQSYRYPYT
101	13A2	Amino acid sequence of a heavy chain CDR1 region	EYVMH
102	13A2	Amino acid sequence of a heavy chain CDR2 region	GINPNTGGTTYNQKFEG
103	13A2	Amino acid sequence of a heavy chain CDR3 region	RMPFGISAMDY
104	13A2	Amino acid sequence of a light chain CDR1 region	SASSSVSYIH
105	13A2	Amino acid sequence of a light chain CDR2 region	ETSKLAS
106	13A2	Amino acid sequence of a light chain CDR3 region	HQWISYPPT
107	14B3	Amino acid sequence of a heavy chain CDR1 region	DHAIH
108	14B3	Amino acid sequence of a heavy chain CDR2 region	VINTYSGNTNFNQKLKG
109	14B3	Amino acid sequence of a heavy chain CDR3 region	GAHSSGYPYWFVDV
110	14B3	Amino acid sequence of a light chain CDR1 region	RASSSVSYM
111	14B3	Amino acid sequence of a light chain CDR2 region	ATSNLAS
112	14B3	Amino acid sequence of a light chain CDR3 region	QQWSTNPFT

Example 5

ADCC Activity Testing of Hybridoma Monoclonal Antibodies

[0071] ADCC (antibody-dependent cell-mediated cytotoxicity) activity of the antibodies was tested using the reporter gene method: 293T-B7H3, PC9 or DLD1 tumor cells were used as target cells, and Jurkat-mCD16.2-NF- κ B or Jurkat-hCD16-NF- κ B cells were used as effector cells. The activation signal of transcription factor NF- κ B mediated by mCD16.2 or hCD16 was used to screen ADCC-positive clones. 50 μ L of hybridoma antibodies that were positive in

the above ELISA and FACS tests were mixed with 25 μ L of 293T-B7H3, PC9 or DLD1 tumor cells (7.5×10^4 cells/well), a resulting mixture was added to a 96-well plate, and then 25 μ L of Jurkat-mCD6. 2-NF- κ B cells (2.5×10^4 cells/well) were added, mixed well, and incubated in an incubator for 4 hours at 37° C. 25 μ L of preheated Bright-Glo solution (Promega, Cat. No.: E2620) was added to each well and left standing for 3 minutes at room temperature in the dark, and the luminescence (Luminescence) signal value of each sample was measured by using a microplate reader (Tecan Spark).

[0072] Among hybridoma antibodies, mouse IgG1 antibodies had no ADCC activity. After gene sequencing, the

inventors confirmed that among the 14 hybridoma antibodies, 9 hybridoma antibodies were based on mouse IgG1, and 5 hybridoma antibodies were based on IgG2b or IgG2a (see Table 3). FIG. 6a shows ADCC reporter gene test results of hybridoma antibodies by using Jurkat-B7H3 as target cells. As expected, 5 hybridoma antibodies based on IgG2b or IgG2a had ADCC activity, while 9 hybridoma antibodies based on mouse IgG1 were inactive. Therefore, as shown in FIG. 6b, for 9 hybridoma antibodies based on mouse IgG1, the inventors conducted Fc replacement and turned these antibodies into chimeric antibodies based on human IgG1. FIG. 7 and FIG. 8 reflect the ADCC reporter gene test results by using PC9 or DLD1 tumor cells as targeted cells. The antibody was in the form of a chimeric antibody based on human IgG1. Based on the test results of ADCC reporter genes, antibodies with relatively strong ADCC activity included 9C8, 5B6, 7C9, F5, 7E6, 2A9, and 4F11.

with a framework sequence of the human germline antibodies. A structural model of the murine antibodies was created to compare amino acids at each site within the framework regions of both humanized and murine antibodies. If the spatial structure of the CDR region was not damaged or changed when a human amino acid sequence was used at a specific site in the framework, the human amino acid sequence was used at the specific site, otherwise, the corresponding murine sequence was used at the site (i.e., a reverse mutation to the murine sequence).
[0075] Based on structural simulations, Ala at the 24th position of a humanized heavy chain of the 2A9 antibody was reversely mutated to Thr, Met at the 48th position was reversely mutated to Ile, Val at the 67th position was reversely mutated to Ala, Met at the 69th position was reversely mutated to Leu, Arg at the 71rd position was reversely mutated to Val, and Thr at the 73rd position was

TABLE 3

Fc Types of B7H3 Hybridoma Antibodies														
Abs	2A9	13A2	4F11	9C8	5B6	7E6	7F5	2F7	7C9	2E10	7B7	14B3	2F12	15A2
Fc	IgG2b	IgG2b	IgG1	IgG1	IgG1	IgG1	IgG1	IgG2b	IgG1	IgG2a	IgG2b	IgG1	IgG1	IgG1

Example 6

Humanizing B7H3 Antibodies 2A9, 4F11, 9C8, 15A2, 5B6 and 7B7

[0073] Considering the various characteristics of hybridoma antibodies, six B7H3 antibodies, namely 2A9, 4F11, 9C8, 15A2, 5B6, and 7B7, were selected for humanization.
[0074] Humanization of these B7H3 antibodies was achieved through the complementary-determining region grafting method. Firstly, IMG T database was queried for human germline antibody (germline antibody) sequences with the maximum sequence homology to the light chain and heavy chain variable regions of murine-derived antibodies 2A9, 4F11, 9C8, 15A2, 5B6 and 7B7. Germline IGKV3-11*01 was selected for humanization of the light chain variable region of the antibody 2A9, and IGHV1-2*02 was selected for humanization of the heavy chain variable region. Germline IGKV3-11*01 was selected for humanization of the light chain variable region of the antibody 4F11, and IGHV1-2*02 was selected for humanization of the heavy chain variable region. Germline IGKV3-11*01 was selected for humanization of the light chain variable region of the antibody 15A2, and IGHV1-8*01 was selected for humanization of the heavy chain variable region. Germline IGKV1-39*01 was selected for humanization of the light chain variable region of the antibody 5B6, and IGHV1-69*02 was selected for humanization of the heavy chain variable region. Germline IGKV1-39*01 was selected for humanization of the light chain variable region of the antibody 7B7, and IGHV3-21*01 was selected for humanization of the heavy chain variable region. The CDR region of the murine-derived antibodies was retained, and the framework (framework) sequence of the murine-derived antibodies was replaced

reversely mutated to Lys. Leu at 46th position of a humanized light chain of 2A9 antibody was reversely mutated to Arg, Leu at the 47th position was reversely mutated to Trp, Ile at the 48th position was reversely mutated to Val, and Phe at the 71st position was reversely mutated to Tyr. Ala at the 24th position of a humanized heavy chain of the antibody 4F11 was reversely mutated to Thr, Met at the 48th position was reversely mutated to Ile, Val at the 67th position was reversely mutated to Ala, Met at the 69th position was reversely mutated to Leu, Arg at the 71st position was reversely mutated to Val, and Thr at the 73rd position was reversely mutated to Lys. Leu at 46th position of a humanized light chain of 4F11 antibody was reversely mutated to Arg, Leu at the 47th position was reversely mutated to Trp, Ile at the 48th position was reversely mutated to Val, and Phe at the 71st position was reversely mutated to Tyr. Met at the 48th position of a humanized heavy chain of the antibody 9C8 was reversely mutated to Ile, Val at the 67th position was reversely mutated to Ala, Met at the 69th position was reversely mutated to Leu, Arg at the 71st position was reversely mutated to Val, and Thr at the 73rd position was reversely mutated to Lys. Ile at 2nd position of a humanized light chain of 9C8 antibody was reversely mutated to Thr, Leu at the 46th position was reversely mutated to Arg, Leu at the 47th position was reversely mutated to Trp, and Phe at the 71st position was reversely mutated to Tyr. Met at the 48th position of a humanized heavy chain of the antibody 15A2 was reversely mutated to Ile, Val at the 67th position was reversely mutated to Ala, Met at the 69th position was reversely mutated to Leu, Arg at the 71st position was reversely mutated to Ala, and Thr at the 73rd position was reversely mutated to Lys. Tyr at 36th position of a humanized light chain of 15A2 antibody was reversely mutated to Phe, Leu at the 47th position was reversely mutated to Trp, Tyr at the 49th position was reversely mutated to His, Ile at the 58th position was reversely mutated to Phe, and Phe at the 71st position was reversely mutated to Tyr. Gly at the 27th position of a humanized heavy chain of 5B6 antibody was

reversely mutated to Tyr, Ser at the 30th position was reversely mutated to Ile, Met at the 48th position was reversely mutated to Ile, Val at the 67th position was reversely mutated to Ala, and Ile at the 69th position was reversely mutated to Leu. Leu at 46th position of a humanized light chain of 5B6 antibody was reversely mutated to Arg, Leu at the 47th position was reversely mutated to Pro, Gly at the 66th position was reversely mutated to Ala, Phe at the 71st position was reversely mutated to Tyr, and Phe at the 98th position was reversely mutated to Ile. Ser at the 49th position of a humanized heavy chain of 7B7 antibody was reversely mutated to Ala. Ile at 2nd position of a humanized light chain of 7B7 antibody was reversely mutated to Ser, and Ile at the 48th position was reversely mutated to Val.

[0076] The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody are SEQ ID NO: 113 and SEQ ID NO: 114, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 4F11 are SEQ ID NO: 115 and SEQ ID NO: 116, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 9C8 are SEQ ID NO: 117 and SEQ ID NO: 118,

respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 15A2 are SEQ ID NO: 119 and SEQ ID NO: 120, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 5B6 are SEQ ID NO: 121 and SEQ ID NO: 122, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 7B7 are SEQ ID NO: 123 and SEQ ID NO: 124, respectively. The above humanized antibodies were constructed as the IgG1 subtype. The amino acid sequence information of the variable regions of the heavy and light chains of humanized antibodies is shown in Table 4.

[0077] Nucleic acid sequences encoding the light and heavy chains of humanized antibodies 2A9, 4F11, 9C8, 15A2, 5B6 and 7B7 were synthesized and inserted into expression vector pcDNA3.1. 200 mL of 293 cells (cell density: 1×10⁶/mL) were co-transfected with 0.1 mg of antibody light chain expression plasmids and 0.1 mg of antibody heavy chain expression plasmids, and cultured while shaken in a shake flask at 37° C. for 6 days, and the supernatant was collected by centrifugation. Protein A was used to purify humanized antibodies, and the purified humanized antibodies were tested for activity.

TABLE 4

Sequence Listing of Humanized B7H3 Antibodies			
SEQ ID NO.	Humanized Antibodies	Description	Amino acid sequence
113	hu2A9	Humanized heavy chain	QVQLVQSGAEVKKPGASVKVCKTS GYTFTEYIMHWVRQAPGGLEWIGGI NPNSGGTTYNEKFKGRATLTVDKSIST AYMELSLRLSDDTAVYYCARRMPYW YFDVWGQGTMTVTSS
114	hu2A9	Humanized light chain	EIVLTQSPATLSLSPGERATLSCSASS SVSYIHWHYQQKPGQAPRRWVYDTSK LASGIPARFSGSGSDTYTLTISLEP EDFAVYYCQQWVSNPLTFGQGTKLEIK
115	hu4F11	Humanized heavy chain	QVQLVQSGAEVKKPGASVKVCKTS GYTFTEYIMHWVRQAPGGLEWIGGI NPNTGGITYNQKFKDRATLTVDKSIST AYMELSLRLSDDTAVYYCARAGGNFL WYFDVWGQGTMTVTSS
116	hu4F11	Humanized light chain	EIVLTQSPATLSLSPGERATLSCSASS SVSYMHWHYQQKPGQAPRRWLYDTS KLASGIPARFSGSGSDTYTLTISLE PEDFAVYYCQQWNSNPLTFGQGTKLEIK
117	hu9C8	Humanized heavy chain	QVQLVQSGAEVKKPGASVKVCKAS GYTFTEYTIHWVRQAPGGLEWIGGI NPNSGGTTYNLKFKGRATLTVDKSTS TVYMELSSRLSDDTAVYYCARKEGFF TTDYWGQGTMTVTSS
118	hu9C8	Humanized light chain	ETVLTQSPATLSLSPGERATLSCSASS SLSYIHWHYQQKPGQAPRRWIYDTSNL ATGIPARFSGSGSDTYTLTISLEPE DFAVYYCQQWHYNPPTFGGKTKVEIK

TABLE 4-continued

Sequence Listing of Humanized B7H3 Antibodies			
SEQ	Description		
ID NO.	Humanized Antibodies	Sequence description	Amino acid sequence
119	hu15A2	Humanized heavy chain	QVQLVQSGAEVKKPGASVKVSCKAS GYTFTSYDINWVRQATGQGLEWIGWI DPGDGSTKYNEKFKGRATLTANKSIST AYMELSSLRSEDTAVYYCVRGFDYW GQGTLLTVSS
120	hu15A2	Humanized light chain	EIVLTQSPATLSLSPGERATLSCRTSSS VSYLHWFOQKPGQAPRLWIHDTSNL ASGFPARFSGSGSDYTLTISLEPE DFAVYYCHQRSSYPWTFGGTKVEIK
121	hu5B6	Humanized heavy chain	QVQLVQSGAEVKKPGSSVKVSCKAS GYTFITYTIHWVRQAPGQGLEWIGYIN PTSDYANFNQKFKDRATLTADKSTSTA YMELSSLRSEDTAVYYCARWGTTVFQ FYFDYWGGTLLTVSS
122	hu5B6	Humanized light chain	EIVLTQSPATLSLSPGERATLSCRASQ SLSYINWYQKPGKAPKPIYATSSSL QSGVPSRFSGSGSDYTLTISLQ EDFATYYCQHWSSDPPTIGGKVEIK
123	hu7B7	Humanized heavy chain	EVQLVESGGGLVKKPGSLRLSCAASG FTFSSYAVSWVRQAPGKGLEWVASIS GGGIYIYYPDSVKGRFTISRDNKNSL YLQMNSLRADTAVYYCARHGGAGY FDYWGGTLLTVSS
124	hu7B7	Humanized light chain	DSQMTQSPSSLSASVGRVTITCRGS ESVHSYLAWYQKPGKAPKLLVYNAL TLQSGVPSRFSGSGSDFTLTISLQ PEDFATYYCQHYGSPPTWTFGGTK VEIK

Example 7

Binding of Humanized B7H3 Antibodies to B7H3

[0078] The binding activity of humanized B7H3 antibody samples to B7H3 protein was assessed using ELISA and FACS, referring to the method detailed in Example 2. The results of humanized antibodies hu2A9, hu4F11, hu5B6, hu9C8, hu15A2 and hu7B7 are shown in Table 5 and Table 6 and FIG. 9 and FIG. 10. Overall, the humanized antibodies according to the present disclosure retained their high antigen affinity from before humanization, particularly notable in hu15A2, hu7B7, and hu5B6.

TABLE 5

Determination of Binding of Humanized B7H3 Antibodies to B7H3-mFc Protein via ELSIA						
EC ₅₀	Antibody					
	hu2A9	hu4F11	hu5B6	hu9C8	hu15A2	hu7B7
(ng/mL)	17.49	40.55	12.73	19.90	3.84	6.35

TABLE 6

Determination of Binding of Humanized B7H3 Antibodies to Cell Surface B7H3 via FACS						
EC ₅₀	Antibody					
	hu2A9	hu4F11	hu5B6	hu9C8	hu15A2	hu7B7
(ng/mL)	140.4	126.7	62.05	79.0	14.3	11.34

Example 8

Detection of Cellular Internalization of Humanized B7H3 Antibodies

[0079] For a specific measurement method, see Example 3. The results are shown in Table 7 and FIG. 11, showing that the humanized B7H3 antibodies had significant cellular internalization by PC9 cells. Compared with the positive control groups (MGA018 from MacroGenics and 8H9 from MSKCC), the cellular internalization of the antibody in the present disclosure was much stronger than that of 8H9. In addition, hu15A2 and hu7B7 also outperformed MGC018.

TABLE 7

Cellular Internalization of Humanized B7H3 Antibodies by PC9 Tumor Cells								
EC ₅₀	Antibody							
	hu2A9	hu4F11	Hu5B6	hu9C8	hu15A2	Hu7B7	MGC018	8H9
(ng/mL)	118.8	88.4	48.88	61.68	29.9.7	16.45	38.16	172.9

Example 9

Cytotoxic Effect of Toxin-Conjugated Humanized B7H3 Antibodies (B7H3-ADC) on Calu-6 Tumor Cells

[0080] Vc-MMAE was conjugated to the B7H3 antibodies in a chemical conjugation method. TCEP was used to reduce B7H3 antibodies in a molar ratio of 3:1, and a resulting mixture was incubated at 37° C. for 45 minutes and subjected to ultrafiltration to remove TCE. The antibodies were resuspended in 0.5 mL of PBS, Vc-MMAE was added to the B7H3 antibodies in a mole ratio of 6 to 1, a resulting mixture was incubated at 4° C. for 120 minutes and subjected to ultrafiltration to remove unconjugated Vc-MMAE, and the antibody concentration was measured for later use. 100 μ L of PC9 (5000 cells per well) and 100 μ L serially diluted B7H3-ADC were added to a 96-well plate. After incubation at 37° C. for 5 days, x μ L of CellTiter-Glo was added to each well, fluorescence intensity was measured, and an inhibition ability of the antibody against tumor cell proliferation was calculated. The results are shown in FIG. 12 and Table 8. Humanized B7H3 antibodies conjugated with MMAE toxin had a significant toxic effect on Calu-6 tumor cells, and was more active than the positive control group (MGC018-MMAE) from MacroGenics.

TABLE 8

Cytotoxic Effect of Conjugate of Humanized B7H3 Antibodies and MMAE on Calu-6 Tumor Cells							
EC ₅₀	Antibody						
	hu7B7-MMAE	hu15A2-MMAE	hu5B6-MMAE	hu2A9-MMAE	hu4F11-MMAE	hu9C8-MMAE	MGC018-MMAE
(pM)	4.877	5.003	4.909	11.45	8.295	16.82	23.37

Example 10

Detection of ADCC Effect of Humanized B7H3 Antibody

[0081] For a specific measurement method, see Example 5. The results are shown in Table 9 and FIG. 13, showing that the humanized B7H3 antibodies had significant ADCC activity. MGA271 (MG-Ab) is an antibody from MacroGenics. Compared with MGA271 (MG-Ab), humanized 9C8, 5B6, 2A9 and 4F11 antibodies had lower EC₅₀ values, indicating better ADCC activity.

TABLE 9

ADCC Effect of Humanized B7H3 Antibody						
Internalization	Antibody					MGA271
	EC ₅₀	hu2A9	hu4F11	Hu5B6	hu9C8	
Jurkat-B7H3	EC ₅₀	43.03	54.79	26.72	32.68	116
PC9	(ng/mL)	88.72	190.4	74.50	70.86	295.3

Example 11

ADCC Cytotoxic Effect of NK92MI-hCD16 Cells on Jurkat-B7H3 and PC9 Cells Mediated by Humanized B7H3 Antibodies

[0082] NK92MI-hCD16 was used to simulate NK cells to evaluate tumor-killing activity of B7H3 antibodies. 25 μ L of PC9 (0.5×10^5 cells per well) stained with CFSE and 25 μ L of NK92MI-hCD16 (1×10^5 cells per well) were mixed, and 50 μ L of serially diluted B7H3 antibodies was added. After the 96-well plate was incubated for 4 hours at 37° C., 5 μ L of 7-AAD was added to each well and the 96-well plate was incubated at room temperature for 10 minutes. The 96-well plate was centrifuged to remove the supernatant and resuspended. A flow cytometer was used to detect a ratio of cells positive for both 7-AAD and CFSE to CFSE-positive cells to obtain the B7H3 antibody-mediated NK cell killing activity. The results are shown in FIG. 14. Humanized B7H3 antibodies 9C8, 5B6, 2A9 and 4F11 could significantly cause an ADCC killing effect of NK92MI-hCD16 cells on PC9 tumor cells.

Example 12

Promoting Effect of Humanized B7H3 Antibodies on Secretion of Cytokines by Human PBMC Cells

[0083] For a specific measurement method, see Example 3. The results are shown in FIG. 15. Humanized B7H3 antibodies 5B6, 2A9 and 4F11 had a significant promoting effect on secretion of IFN- γ by human PBMC cells.

[0084] The foregoing embodiments are only preferred embodiments of the present disclosure and cannot be construed as a limitation on the protection scope of the present disclosure. That is, any simple equivalent changes and modifications made in accordance with the claims of the present disclosure and content of the utility model still fall within the protection scope of the present disclosure.

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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Leu Val Lys Leu Ser Cys Lys Thr Ser Gly Phe Asn Ile Lys Ala Tyr
20 25 30

Tyr Ile His Trp Val Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile
35 40 45

Gly Trp Ile Asp Pro Glu Asn Gly Asn Thr Phe Tyr Asp Pro Lys Phe
50 55 60

Gln Gly Lys Ala Ser Ile Thr Ala Asp Thr Ser Ser Asn Thr Ala Tyr
65 70 75 80

Leu Gln Leu Ser Ser Leu Thr Ser Glu Asp Ala Ala Val Tyr Tyr Cys
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Thr Thr Leu Thr Val Ser Ser
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Asp Ile Val Met Thr Gln Ala Ala Phe Ser Asn Pro Val Thr Leu Gly
1 5 10 15

Thr Ser Ala Ser Ile Ser Cys Arg Ser Ser Lys Ser Leu Leu His Ser
20 25 30

Asn Gly Ile Thr Tyr Leu Tyr Trp Tyr Leu Gln Lys Ser Gly Gln Ser
35 40 45

Pro Gln Leu Leu Ile Tyr Gln Met Ser Asn Leu Ala Ser Gly Val Pro
50 55 60

Asp Arg Phe Ser Ser Ser Gly Ser Gly Thr Asp Phe Thr Leu Arg Ile
65 70 75 80

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85 90 95

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<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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 1 5 10 15

Ser Val Lys Ile Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Glu Tyr
 20 25 30

Thr Ile His Trp Val Lys Gln Ser His Gly Lys Ser Leu Glu Trp Ile
 35 40 45

Gly Gly Ile Asn Pro Asn Ser Gly Gly Thr Thr Tyr Asn Leu Lys Phe
 50 55 60

Lys Gly Lys Ala Thr Leu Thr Val Asp Lys Ser Ser Ser Thr Val Tyr
 65 70 75 80

Met Glu Leu Arg Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Lys Glu Gly Phe Phe Thr Thr Asp Tyr Trp Gly Gln Gly Thr
 100 105 110

Thr Leu Thr Val Ser Ser
 115

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 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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 1 5 10 15

Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Leu Ser Tyr Ile
 20 25 30

His Trp Tyr Arg Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp Ile Tyr
 35 40 45

Asp Thr Ser Asn Leu Ala Thr Gly Val Pro Val Arg Phe Ser Gly Ser
 50 55 60

Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu Ala Glu
 65 70 75 80

Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp His Tyr Asn Pro Pro Thr
 85 90 95

Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105

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 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 5

Gln Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Ala Arg Pro Gly Ala
 1 5 10 15

Ser Val Lys Met Ser Cys Lys Thr Ser Gly Tyr Thr Phe Ile Thr Tyr
 20 25 30

Gln	Ile	Val	Leu	Ser	Gln	Ser	Pro	Ala	Ile	Leu	Ser	Ala	Ser	Pro	Gly
1				5					10					15	
Glu	Lys	Val	Thr	Met	Thr	Cys	Arg	Ala	Ser	Gln	Ser	Leu	Ser	Tyr	Ile
			20					25					30		
Asn	Trp	Tyr	Gln	Leu	Lys	Pro	Gly	Ser	Ser	Pro	Lys	Pro	Trp	Ile	Tyr
		35					40					45			
Ala	Thr	Ser	Ser	Leu	Gln	Ser	Gly	Val	Pro	Ala	Arg	Phe	Ser	Gly	Ser
	50					55					60				
Ala	Ser	Gly	Thr	Ser	Tyr	Ser	Leu	Thr	Ile	Ser	Arg	Val	Glu	Ala	Glu
65					70					75					80
Asp	Ala	Ala	Thr	Tyr	Phe	Cys	Gln	His	Trp	Ser	Ser	Asp	Pro	Pro	Thr
			85						90					95	
Ile	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Lys						
			100					105							

Glu	Val	Gln	Leu	Gln	Gln	Ser	Gly	Thr	Glu	Leu	Val	Arg	Pro	Gly	Ala
1				5					10					15	
Leu	Val	Lys	Leu	Ser	Cys	Lys	Ala	Ser	Ala	Leu	Asn	Ile	Lys	Asp	Phe
			20					25					30		
Tyr	Met	His	Trp	Val	Lys	Gln	Arg	Pro	Glu	Arg	Gly	Leu	Glu	Trp	Ile
		35					40					45			
Gly	Trp	Ile	Asp	Pro	Asp	Asn	Gly	Asn	Thr	Ile	Tyr	Asp	Pro	Lys	Phe
	50					55					60				
Gln	Gly	Lys	Ala	Ser	Ile	Thr	Ala	Asp	Ile	Ser	Ser	Ser	Thr	Ala	Tyr
65					70				75						80

-continued

Leu Gln Leu Ser Ser Leu Thr Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Ser Ser Thr Arg Gln Arg Pro Phe Asp Tyr Trp Gly Gln Gly
100 105 110

Thr Thr Leu Thr Val Ser Ser
115

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Asp Ile Val Met Thr Gln Ser His Lys Val Met Ser Thr Ser Val Gly
1 5 10 15

Asp Arg Val Ser Ile Thr Cys Lys Ala Ser Gln Asp Val Arg Thr Gly
20 25 30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Lys Leu Leu Ile
35 40 45

Tyr Ser Ala Phe Tyr Arg Tyr Thr Gly Val Pro Asp Arg Phe Thr Val
50 55 60

Ser Gly Ser Gly Ala Asp Phe Thr Phe Thr Ile Thr Ser Val Gln Ala
65 70 75 80

Glu Asp Leu Ala Val Tyr Tyr Cys Gln Gln Tyr Tyr Ser Ile Pro Trp
85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 9

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Ile Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Glu Tyr
20 25 30

Ile Met His Trp Val Lys Gln Ser Arg Gly Lys Ser Leu Glu Trp Ile
35 40 45

Gly Gly Ile Asn Pro Asn Ser Gly Gly Thr Thr Tyr Asn Glu Lys Phe
50 55 60

Lys Gly Lys Ala Thr Leu Thr Val Asp Lys Ser Ala Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Arg Ser Leu Thr Ser Glu Asp Ser Ala Ile Tyr Tyr Cys
85 90 95

Ala Arg Arg Met Pro Tyr Trp Tyr Phe Asp Val Trp Gly Ala Gly Thr
100 105 110

Thr Val Thr Val Ser Ser
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Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Pro Gly
1 5 10 15
Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr Ile
20 25 30
His Trp Tyr Arg Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp Val Tyr
35 40 45
Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ala Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu Ala Glu
65 70 75 80
Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Val Ser Asn Pro Leu Thr
85 90 95
Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
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Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
1 5 10 15
Ser Val Lys Leu Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Glu Tyr
20 25 30
Ile Met His Trp Val Lys Gln Ser His Gly Lys Ser Leu Glu Trp Ile
35 40 45
Gly Gly Ile Asn Pro Asn Thr Gly Gly Ile Thr Tyr Asn Gln Lys Phe
50 55 60
Lys Asp Lys Ala Thr Leu Thr Val Asp Lys Ser Ser Ser Thr Ala Tyr
65 70 75 80
Met Glu Leu Arg Ser Leu Thr Ser Asp Asp Ser Ala Val Tyr Phe Ser
85 90 95
Ala Arg Ala Gly Gly Asn Phe Leu Trp Tyr Phe Asp Val Trp Gly Ala
100 105 110
Gly Thr Thr Val Thr Val Ser Ser
115 120

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Gln Ile Val Leu Thr Gln Ser Pro Ala Thr Met Ser Ala Ser Pro Gly
 1 5 10 15
 Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr Met
 20 25 30
 His Trp Tyr Gln Gln Lys Ser Asp Met Ser Pro Lys Arg Trp Leu Tyr
 35 40 45
 Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ala Arg Phe Ser Gly Ser
 50 55 60
 Gly Ser Gly Thr Ser Tyr Ser Leu Ala Ile Ser Ser Met Glu Ala Glu
 65 70 75 80
 Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Asn Ser Asn Pro Leu Thr
 85 90 95
 Phe Gly Thr Gly Thr Lys Leu Glu Leu Lys
 100 105

<210> SEQ ID NO 13

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<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 13

Gln Val Gln Leu Gln Gln Ser Gly Pro Glu Val Val Lys Pro Gly Ala
 1 5 10 15
 Leu Val Arg Ile Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr
 20 25 30
 Asp Ile Asn Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Trp Ile Asp Pro Gly Asp Gly Ser Thr Lys Tyr Asn Glu Lys Phe
 50 55 60
 Lys Gly Lys Ala Thr Leu Thr Ala Asp Lys Ser Ser Ser Thr Ala Tyr
 65 70 75 80
 Met Gln Phe Ser Ser Leu Thr Ser Glu Asn Ser Ala Val Tyr Phe Cys
 85 90 95
 Val Arg Arg Gly Phe Asp Tyr Trp Gly Gln Gly Thr Thr Leu Thr Val
 100 105 110
 Ser Ser

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Gln Ile Leu Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Pro Gly
 1 5 10 15
 Glu Lys Val Thr Met Thr Cys Arg Thr Ser Ser Ser Val Ser Tyr Leu
 20 25 30
 His Trp Phe Gln Gln Lys Pro Gly Ser Ser Pro Lys Leu Trp Ile His

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35	40	45
Asp Thr Ser Asn Leu Ala Ser Gly Phe Pro Ala Arg Phe Ser Gly Ser		
50	55	60
Gly Ser Gly Thr Ser Tyr Ser Leu Ile Ile Ser Ser Met Glu Ala Glu		
65	70	75 80
Asp Ala Ala Thr Tyr Tyr Cys His Gln Arg Ser Ser Tyr Pro Trp Thr		
	85	90 95
Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys		
100	105	

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 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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Glu Val Lys Leu Val Glu Ser Gly Gly Gly Leu Val Arg Pro Gly Gly
1 5 10 15
Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30
Ala Val Ser Trp Val Arg Gln Thr Pro Glu Lys Arg Leu Glu Trp Val
35 40 45
Ala Ser Ile Ser Gly Gly Gly Ile Tyr Ile Tyr Tyr Pro Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Asn Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Thr Pro Asp Asp Thr Ala Leu Tyr Tyr Cys
85 90 95
Ala Arg His Gly Gly Ala Gly Tyr Phe Asp Tyr Trp Gly Gln Gly Thr
100 105 110
Thr Leu Thr Val Ser Ser
115

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Asp Ser Leu Met Thr Gln Ser Pro Ala Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Glu Thr Val Thr Ile Thr Cys Arg Gly Ser Glu Ser Val His Ser Tyr
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Gln Gly Lys Ser Pro Gln Leu Leu Val
35 40 45
Tyr Asn Ala Lys Thr Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Gln Phe Ser Leu Arg Ile Asn Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Gly Thr Tyr Tyr Cys Gln Gln His Tyr Gly Ser Pro Pro

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	85		90		95
Trp	Thr	Phe	Gly	Gly	Gly
			100	Thr	Ile
				Leu	Glu
				Ile	Lys
				105	

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 <220> FEATURE:
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<400> SEQUENCE: 17

Gln	Val	Gln	Leu	Gln	Gln	Ser	Gly	Pro	Glu	Leu	Val	Arg	Pro	Gly	Val
1			5						10					15	
Ser	Val	Lys	Ile	Ser	Cys	Lys	Gly	Ser	Gly	Tyr	Thr	Phe	Thr	Asp	Tyr
		20						25					30		
Ala	Met	His	Trp	Val	Lys	Gln	Ser	His	Ala	Lys	Ser	Leu	Glu	Trp	Ile
		35					40					45			
Gly	Val	Ile	Asn	Ser	Tyr	Tyr	Gly	Asn	Thr	His	Tyr	Asn	Gln	Lys	Phe
	50					55					60				
Lys	Gly	Lys	Ala	Thr	Met	Thr	Val	Asp	Lys	Ser	Ser	Ser	Thr	Ala	Tyr
65					70				75					80	
Met	Glu	Leu	Ala	Arg	Leu	Thr	Ser	Glu	Asp	Ser	Ala	Ile	Tyr	Tyr	Cys
			85					90					95		
Ala	Arg	Ala	Gly	Gly	Tyr	Ser	Tyr	Gly	Ser	Ser	Tyr	Arg	Pro	Asp	Arg
		100						105					110		
Phe	Ala	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Ile	Val	Ser	Ala		
		115					120					125			

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 <212> TYPE: PRT
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<400> SEQUENCE: 18

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ala	Ser	Gln	Ser	Ala	Ser	Leu	Gly
1			5						10					15	
Glu	Ser	Val	Thr	Ile	Thr	Cys	Leu	Ala	Ser	Gln	Thr	Ile	Gly	Thr	Trp
		20						25					30		
Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ser	Pro	Gln	Leu	Leu	Ile
		35					40					45			
Tyr	Ala	Ala	Thr	Asn	Leu	Ala	Asp	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50				55					60					
Ser	Gly	Ser	Gly	Thr	Lys	Phe	Ser	Phe	Lys	Ile	Ser	Ser	Leu	Gln	Ala
65				70					75					80	
Glu	Asp	Phe	Val	Ser	Tyr	Tyr	Cys	Gln	Gln	Leu	Tyr	Ser	Thr	Pro	Tyr
			85					90					95		
Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Lys					
		100					105								

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Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
1 5 10 15
Ser Val Lys Ile Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Glu Tyr
20 25 30
Thr Met His Trp Val Lys Gln Ser His Gly Lys Asn Leu Glu Trp Ile
35 40 45
Gly Gly Ile Asn Thr Asp Asn Gly Gly Thr Thr Tyr Ser Gln Lys Phe
50 55 60
Lys Asp Lys Ala Thr Leu Thr Val Asp Arg Ser Ser Ser Thr Ala Tyr
65 70 75 80
Met Val Leu Arg Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
85 90 95
Ala Arg Ser Phe Asn Asn Tyr Gly Pro Ala Trp Phe Ala Tyr Trp Gly
100 105 110
Gln Gly Ala Leu Val Thr Val Ser Ala
115 120

<210> SEQ ID NO 20
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 20

Asp Ile Val Met Thr Gln Ser His Lys Phe Met Ser Thr Ser Ile Gly
1 5 10 15
Asp Arg Val Ser Ile Thr Cys Lys Ala Ser Gln Asp Val Arg Ser Ala
20 25 30
Val Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ser Pro Lys Leu Leu Ile
35 40 45
Tyr Trp Ala Ser Thr Arg His Thr Gly Val Pro Ala Arg Phe Thr Gly
50 55 60
Ser Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Asn Asn Val Gln Ser
65 70 75 80
Glu Asp Leu Ala Gly Tyr Phe Cys Gln Gln Tyr Ser Ser Tyr Phe Thr
85 90 95
Phe Gly Gly Gly Thr Lys Leu Glu Lys Lys
100 105

<210> SEQ ID NO 21
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<212> TYPE: PRT
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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Ser Gly Ala
1 5 10 15

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Ser Val Asn Ile Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Glu Asn
      20                      25                      30

Thr Met His Trp Val Lys Leu Ser His Gly Lys Ser Leu Glu Trp Ile
      35                      40                      45

Gly Gly Ile Asn Pro Asn Asn Gly Gly Thr Thr Tyr Asn Gln Lys Phe
      50                      55                      60

Lys Gly Lys Ala Thr Leu Thr Val Asp Lys Ser Ser Ser Thr Ala Tyr
      65                      70                      75                      80

Met Glu Leu Arg Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
      85                      90                      95

Ala Arg Arg Gly Arg Trp Pro Pro Asp Trp Tyr Phe Asp Val Trp Gly
      100                     105                     110

Ala Gly Thr Thr Val Thr Val Ser Ser
      115                      120

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<210> SEQ ID NO 22
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<400> SEQUENCE: 22

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Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Pro Gly
1      5                      10                      15

Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Leu Ser Tyr Met
      20                      25                      30

His Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp Ile Tyr
      35                      40                      45

Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ala Arg Phe Ser Gly Ser
      50                      55                      60

Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu Ala Glu
      65                      70                      75                      80

Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro Leu Thr
      85                      90                      95

Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
      100                     105

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<210> SEQ ID NO 23
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<212> TYPE: PRT
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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 23

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Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Lys Pro Gly Ala
1      5                      10                      15

Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr
      20                      25                      30

Tyr Ile Tyr Trp Leu Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
      35                      40                      45

Gly Gly Ile Lys Pro Ser Asn Gly Asp Thr Asn Phe Asn Ala Lys Phe
      50                      55                      60

```

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Met Asn Lys Ala Thr Leu Thr Val Asp Lys Ser Ser Ser Thr Ala Tyr
65 70 75 80

Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Ile Tyr Tyr Cys
85 90 95

Thr Arg Phe Phe Tyr Gly Ser Pro Tyr Asp Phe Asp Val Trp Gly Ala
100 105 110

Gly Thr Thr Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 24

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 24

Asn Ile Val Met Thr Gln Ser Pro Lys Ser Met Ser Met Ala Val Gly
1 5 10 15

Glu Arg Val Thr Leu Thr Cys Lys Ala Ser Glu Asn Val Asp Thr His
20 25 30

Val Ser Trp Tyr Gln Gln Lys Pro Glu Gln Ser Pro Thr Leu Leu Met
35 40 45

Tyr Gly Thr Ser Asn Arg Phe Thr Gly Val Pro Asp Arg Phe Thr Gly
50 55 60

Ser Gly Ser Ala Thr Asp Phe Thr Leu Thr Ile Ser Ser Val Gln Ala
65 70 75 80

Glu Asp Leu Ala Asp Tyr His Cys Gly Gln Ser Tyr Arg Tyr Pro Tyr
85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 25

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 25

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Ile Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Glu Tyr
20 25 30

Val Met His Trp Val Lys Gln Arg His Gly Gln Ser Leu Glu Trp Ile
35 40 45

Gly Gly Ile Asn Pro Asn Thr Gly Gly Thr Thr Tyr Asn Gln Lys Phe
50 55 60

Glu Gly Lys Ala Thr Leu Thr Val Asp Lys Ser Ser Ser Ser Ala Tyr
65 70 75 80

Met Glu Phe Arg Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Arg Met Pro Phe Gly Ile Ser Ala Met Asp Tyr Trp Gly Gln
100 105 110

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Gly Thr Ser Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 26
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 26

Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Pro Gly
1 5 10 15
Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr Ile
20 25 30
His Trp Phe Gln Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp Ile Ser
35 40 45
Glu Thr Ser Lys Leu Ala Ser Gly Val Pro Ala Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Gly Ala Glu
65 70 75 80
Asp Ala Ala Thr Tyr Tyr Cys His Gln Trp Ile Ser Tyr Pro Pro Thr
85 90 95
Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
100 105

<210> SEQ ID NO 27
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 27

Gln Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Arg Pro Gly Val
1 5 10 15
Ser Val Lys Ile Ser Cys Lys Gly Ser Gly Tyr Thr Phe Thr Asp His
20 25 30
Ala Ile His Trp Val Lys Gln Ser His Ala Lys Ser Leu Glu Trp Ile
35 40 45
Gly Val Ile Asn Thr Tyr Ser Gly Asn Thr Asn Phe Asn Gln Lys Leu
50 55 60
Lys Gly Lys Ala Thr Met Thr Val Asp Lys Ser Ser Ser Thr Ala Tyr
65 70 75 80
Met Glu Leu Ala Arg Leu Thr Ser Glu Asp Ser Ala Ile Tyr Tyr Cys
85 90 95
Ala Arg Gly Ala His Ser Ser Gly Tyr Pro Tyr Trp Tyr Phe Asp Val
100 105 110
Trp Gly Ala Gly Thr Thr Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 28
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 28

Gln Ile Val Leu Ser Gln Ser Pro Ala Ile Leu Ser Ala Ser Ala Gly
1 5 10 15
Glu Lys Val Thr Met Thr Cys Arg Ala Ser Ser Ser Val Ser Tyr Met
20 25 30
His Trp Tyr Gln Gln Lys Pro Gly Ser Ser Pro Lys Pro Trp Ile Tyr
35 40 45
Ala Thr Ser Asn Leu Ala Ser Gly Val Pro Ala Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Arg Val Glu Ala Glu
65 70 75 80
Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Thr Asn Pro Phe Thr
85 90 95
Phe Gly Ser Gly Thr Lys Leu Glu Met Lys
100 105

<210> SEQ ID NO 29

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 29

Ala Tyr Tyr Ile His
1 5

<210> SEQ ID NO 30

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 30

Trp Ile Asp Pro Glu Asn Gly Asn Thr Phe Tyr Asp Pro Lys Phe Gln
1 5 10 15
Gly

<210> SEQ ID NO 31

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 31

Ser Ser Thr Arg Leu Arg Pro Phe Asp Tyr
1 5 10

<210> SEQ ID NO 32

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 32

Arg Ser Ser Lys Ser Leu Leu His Ser Asn Gly Ile Thr Tyr Leu Tyr
1 5 10 15

<210> SEQ ID NO 33
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 33

Gln Met Ser Asn Leu Ala Ser
1 5

<210> SEQ ID NO 34
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 34

Ala Gln Asn Leu Glu Leu Tyr Thr
1 5

<210> SEQ ID NO 35
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 35

Glu Tyr Thr Ile His
1 5

<210> SEQ ID NO 36
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 36

Gly Ile Asn Pro Asn Ser Gly Gly Thr Thr Tyr Asn Leu Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 37
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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<400> SEQUENCE: 37

Lys Glu Gly Phe Phe Thr Thr Asp Tyr
1 5

<210> SEQ ID NO 38

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 38

Ser Ala Ser Ser Ser Leu Ser Tyr Ile His
1 5 10

<210> SEQ ID NO 39

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 39

Asp Thr Ser Asn Leu Ala Thr
1 5

<210> SEQ ID NO 40

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 40

Gln Gln Trp His Tyr Asn Pro Pro Thr
1 5

<210> SEQ ID NO 41

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 41

Thr Tyr Thr Ile His
1 5

<210> SEQ ID NO 42

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 42

Tyr Ile Asn Pro Thr Ser Asp Tyr Ala Asn Phe Asn Gln Lys Phe Lys
1 5 10 15

Asp

-continued

<210> SEQ ID NO 43
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 43

Trp Gly Thr Thr Val Phe Gln Phe Tyr Phe Asp Tyr
1 5 10

<210> SEQ ID NO 44
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 44

Arg Ala Ser Gln Ser Leu Ser Tyr Ile Asn
1 5 10

<210> SEQ ID NO 45
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 45

Ala Thr Ser Ser Leu Gln Ser
1 5

<210> SEQ ID NO 46
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 46

Gln His Trp Ser Ser Asp Pro Pro Thr
1 5

<210> SEQ ID NO 47
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 47

Asp Phe Tyr Met His
1 5

<210> SEQ ID NO 48
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 48

Trp Ile Asp Pro Asp Asn Gly Asn Thr Ile Tyr Asp Pro Lys Phe Gln
1 5 10 15

Gly

<210> SEQ ID NO 49
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 49

Ser Ser Thr Arg Gln Arg Pro Phe Asp Tyr
1 5 10

<210> SEQ ID NO 50
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 50

Lys Ala Ser Gln Asp Val Arg Thr Gly Val Ala
1 5 10

<210> SEQ ID NO 51
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 51

Ser Ala Phe Tyr Arg Tyr Thr
1 5

<210> SEQ ID NO 52
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 52

Gln Gln Tyr Tyr Ser Ile Pro Trp Thr
1 5

<210> SEQ ID NO 53
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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<400> SEQUENCE: 53

Glu Tyr Ile Met His
1 5

<210> SEQ ID NO 54

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 54

Gly Ile Asn Pro Asn Ser Gly Gly Thr Thr Tyr Asn Glu Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 55

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 55

Arg Met Pro Tyr Trp Tyr Phe Asp Val
1 5

<210> SEQ ID NO 56

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 56

Ser Ala Ser Ser Ser Val Ser Tyr Ile His
1 5 10

<210> SEQ ID NO 57

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 57

Asp Thr Ser Lys Leu Ala Ser
1 5

<210> SEQ ID NO 58

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 58

Gln Gln Trp Val Ser Asn Pro Leu Thr
1 5

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<210> SEQ ID NO 59
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 59

Glu Tyr Ile Met His
1 5

<210> SEQ ID NO 60
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 60

Gly Ile Asn Pro Asn Thr Gly Gly Ile Thr Tyr Asn Gln Lys Phe Lys
1 5 10 15

Asp

<210> SEQ ID NO 61
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 61

Ala Gly Gly Asn Phe Leu Trp Tyr Phe Asp Val
1 5 10

<210> SEQ ID NO 62
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 62

Ser Ala Ser Ser Ser Val Ser Tyr Met His
1 5 10

<210> SEQ ID NO 63
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 63

Asp Thr Ser Lys Leu Ala Ser
1 5

<210> SEQ ID NO 64
<211> LENGTH: 9

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 64

Gln Gln Trp Asn Ser Asn Pro Leu Thr
1 5

<210> SEQ ID NO 65
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 65

Ser Tyr Asp Ile Asn
1 5

<210> SEQ ID NO 66
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 66

Trp Ile Asp Pro Gly Asp Gly Ser Thr Lys Tyr Asn Glu Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 67
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 67

Arg Gly Phe Asp Tyr
1 5

<210> SEQ ID NO 68
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 68

Arg Thr Ser Ser Ser Val Ser Tyr Leu His
1 5 10

<210> SEQ ID NO 69
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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peptide

<400> SEQUENCE: 69

Asp Thr Ser Asn Leu Ala Ser
1 5

<210> SEQ ID NO 70

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 70

His Gln Arg Ser Ser Tyr Pro Trp Thr
1 5

<210> SEQ ID NO 71

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 71

Ser Tyr Ala Val Ser
1 5

<210> SEQ ID NO 72

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 72

Ser Ile Ser Gly Gly Gly Ile Tyr Ile Tyr Tyr Pro Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 73

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 73

His Gly Gly Ala Gly Tyr Phe Asp Tyr
1 5

<210> SEQ ID NO 74

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 74

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Arg Gly Ser Glu Ser Val His Ser Tyr Leu Ala
1 5 10

<210> SEQ ID NO 75
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 75

Asn Ala Lys Thr Leu Gln Ser
1 5

<210> SEQ ID NO 76
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 76

Gln Gln His Tyr Gly Ser Pro Pro Trp Thr
1 5 10

<210> SEQ ID NO 77
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 77

Asp Tyr Ala Met His
1 5

<210> SEQ ID NO 78
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 78

Val Ile Asn Ser Tyr Tyr Gly Asn Thr His Tyr Asn Gln Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 79
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 79

Ala Gly Gly Tyr Ser Tyr Gly Ser Ser Tyr Arg Pro Asp Arg Phe Ala
1 5 10 15

Tyr

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<210> SEQ ID NO 80
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 80

Leu Ala Ser Gln Thr Ile Gly Thr Trp Leu Ala
1 5 10

<210> SEQ ID NO 81
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 81

Ala Ala Thr Asn Leu Ala Asp
1 5

<210> SEQ ID NO 82
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 82

Gln Gln Leu Tyr Ser Thr Pro Tyr Thr
1 5

<210> SEQ ID NO 83
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 83

Glu Tyr Thr Met His
1 5

<210> SEQ ID NO 84
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 84

Gly Ile Asn Thr Asp Asn Gly Gly Thr Thr Tyr Ser Gln Lys Phe Lys
1 5 10 15

Asp

<210> SEQ ID NO 85
<211> LENGTH: 12

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 85

Ser Phe Asn Asn Tyr Gly Pro Ala Trp Phe Ala Tyr
1 5 10

<210> SEQ ID NO 86
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 86

Lys Ala Ser Gln Asp Val Arg Ser Ala Val Ala
1 5 10

<210> SEQ ID NO 87
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 87

Trp Ala Ser Thr Arg His Thr
1 5

<210> SEQ ID NO 88
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 88

Gln Gln Tyr Ser Ser Tyr Phe Thr
1 5

<210> SEQ ID NO 89
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 89

Glu Asn Thr Met His
1 5

<210> SEQ ID NO 90
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

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<400> SEQUENCE: 90

Gly Ile Asn Pro Asn Asn Gly Gly Thr Thr Tyr Asn Gln Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 91

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 91

Arg Gly Arg Trp Pro Pro Asp Trp Tyr Phe Asp Val
1 5 10

<210> SEQ ID NO 92

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 92

Ser Ala Ser Ser Ser Leu Ser Tyr Met His
1 5 10

<210> SEQ ID NO 93

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 93

Asp Thr Ser Lys Leu Ala Ser
1 5

<210> SEQ ID NO 94

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 94

Gln Gln Trp Ser Ser Asn Pro Leu Thr
1 5

<210> SEQ ID NO 95

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 95

Ser Tyr Tyr Ile Tyr
1 5

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<210> SEQ ID NO 96
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 96

Gly Ile Lys Pro Ser Asn Gly Asp Thr Asn Phe Asn Ala Lys Phe Met
1 5 10 15

Asn

<210> SEQ ID NO 97
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 97

Phe Phe Tyr Gly Ser Pro Tyr Asp Phe Asp Val
1 5 10

<210> SEQ ID NO 98
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 98

Lys Ala Ser Glu Asn Val Asp Thr His Val Ser
1 5 10

<210> SEQ ID NO 99
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 99

Gly Thr Ser Asn Arg Phe Thr
1 5

<210> SEQ ID NO 100
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 100

Gly Gln Ser Tyr Arg Tyr Pro Tyr Thr
1 5

<210> SEQ ID NO 101
<211> LENGTH: 5

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 101

Glu Tyr Val Met His
1 5

<210> SEQ ID NO 102
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 102

Gly Ile Asn Pro Asn Thr Gly Gly Thr Thr Tyr Asn Gln Lys Phe Glu
1 5 10 15

Gly

<210> SEQ ID NO 103
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 103

Arg Met Pro Phe Gly Ile Ser Ala Met Asp Tyr
1 5 10

<210> SEQ ID NO 104
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 104

Ser Ala Ser Ser Ser Val Ser Tyr Ile His
1 5 10

<210> SEQ ID NO 105
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 105

Glu Thr Ser Lys Leu Ala Ser
1 5

<210> SEQ ID NO 106
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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peptide

<400> SEQUENCE: 106

His Gln Trp Ile Ser Tyr Pro Pro Thr
1 5

<210> SEQ ID NO 107

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 107

Asp His Ala Ile His
1 5

<210> SEQ ID NO 108

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 108

Val Ile Asn Thr Tyr Ser Gly Asn Thr Asn Phe Asn Gln Lys Leu Lys
1 5 10 15

Gly

<210> SEQ ID NO 109

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 109

Gly Ala His Ser Ser Gly Tyr Pro Tyr Trp Tyr Phe Asp Val
1 5 10

<210> SEQ ID NO 110

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 110

Arg Ala Ser Ser Ser Val Ser Tyr Met His
1 5 10

<210> SEQ ID NO 111

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 111

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Ala Thr Ser Asn Leu Ala Ser
1 5

<210> SEQ ID NO 112
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 112

Gln Gln Trp Ser Thr Asn Pro Phe Thr
1 5

<210> SEQ ID NO 113
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 113

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Val Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Glu Tyr
20 25 30

Ile Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35 40 45

Gly Gly Ile Asn Pro Asn Ser Gly Gly Thr Thr Tyr Asn Glu Lys Phe
50 55 60

Lys Gly Arg Ala Thr Leu Thr Val Asp Lys Ser Ile Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Arg Met Pro Tyr Trp Tyr Phe Asp Val Trp Gly Gln Gly Thr
100 105 110

Met Val Thr Val Ser Ser
115

<210> SEQ ID NO 114
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 114

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Ser Ala Ser Ser Ser Val Ser Tyr Ile
20 25 30

His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Arg Trp Val Tyr
35 40 45

Asp Thr Ser Lys Leu Ala Ser Gly Ile Pro Ala Arg Phe Ser Gly Ser
50 55 60

Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Glu Pro Glu

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65              70              75              80
Asp Phe Ala Val Tyr Tyr Cys Gln Gln Trp Val Ser Asn Pro Leu Thr
      85              90              95

Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
      100              105

<210> SEQ ID NO 115
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 115

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1          5          10          15
Ser Val Lys Val Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Glu Tyr
      20          25          30
Ile Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
      35          40          45
Gly Gly Ile Asn Pro Asn Thr Gly Gly Ile Thr Tyr Asn Gln Lys Phe
      50          55          60
Lys Asp Arg Ala Thr Leu Thr Val Asp Lys Ser Ile Ser Thr Ala Tyr
      65          70          75          80
Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
      85          90          95
Ala Arg Ala Gly Gly Asn Phe Leu Trp Tyr Phe Asp Val Trp Gly Gln
      100          105          110
Gly Thr Met Val Thr Val Ser Ser
      115          120

<210> SEQ ID NO 116
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 116

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1          5          10          15
Glu Arg Ala Thr Leu Ser Cys Ser Ala Ser Ser Ser Val Ser Tyr Met
      20          25          30
His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Arg Trp Leu Tyr
      35          40          45
Asp Thr Ser Lys Leu Ala Ser Gly Ile Pro Ala Arg Phe Ser Gly Ser
      50          55          60
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Glu Pro Glu
      65          70          75          80
Asp Phe Ala Val Tyr Tyr Cys Gln Gln Trp Asn Ser Asn Pro Leu Thr
      85          90          95
Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
      100          105

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<210> SEQ ID NO 117
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 117

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1 5 10 15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Glu Tyr
20 25 30
Thr Ile His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
35 40 45
Gly Gly Ile Asn Pro Asn Ser Gly Gly Thr Thr Tyr Asn Leu Lys Phe
50 55 60
Lys Gly Arg Ala Thr Leu Thr Val Asp Lys Ser Thr Ser Thr Val Tyr
65 70 75 80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Ala Arg Lys Glu Gly Phe Phe Thr Thr Asp Tyr Trp Gly Gln Gly Thr
100 105 110
Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 118
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 118

Glu Thr Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15
Glu Arg Ala Thr Leu Ser Cys Ser Ala Ser Ser Ser Leu Ser Tyr Ile
20 25 30
His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Arg Trp Ile Tyr
35 40 45
Asp Thr Ser Asn Leu Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Glu Pro Glu
65 70 75 80
Asp Phe Ala Val Tyr Tyr Cys Gln Gln Trp His Tyr Asn Pro Pro Thr
85 90 95
Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 119
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 119

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Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr
 20 25 30
 Asp Ile Asn Trp Val Arg Gln Ala Thr Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Trp Ile Asp Pro Gly Asp Gly Ser Thr Lys Tyr Asn Glu Lys Phe
 50 55 60
 Lys Gly Arg Ala Thr Leu Thr Ala Asn Lys Ser Ile Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Val Arg Arg Gly Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val
 100 105 110

Ser Ser

<210> SEQ ID NO 120
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 120

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Thr Ser Ser Ser Val Ser Tyr Leu
 20 25 30
 His Trp Phe Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Trp Ile His
 35 40 45
 Asp Thr Ser Asn Leu Ala Ser Gly Phe Pro Ala Arg Phe Ser Gly Ser
 50 55 60
 Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Glu Pro Glu
 65 70 75 80
 Asp Phe Ala Val Tyr Tyr Cys His Gln Arg Ser Ser Tyr Pro Trp Thr
 85 90 95
 Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 121
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 121

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Ile Thr Tyr
 20 25 30
 Thr Ile His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

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Gly	Tyr	Ile	Asn	Pro	Thr	Ser	Asp	Tyr	Ala	Asn	Phe	Asn	Gln	Lys	Phe
	50					55					60				

Lys	Asp	Arg	Ala	Thr	Leu	Thr	Ala	Asp	Lys	Ser	Thr	Ser	Thr	Ala	Tyr
65					70				75					80	

Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85					90						95	

Ala	Arg	Trp	Gly	Thr	Thr	Val	Phe	Gln	Phe	Tyr	Phe	Asp	Tyr	Trp	Gly
			100					105					110		

Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser
	115						120	

<210> SEQ ID NO 122

<211> LENGTH: 106

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 122

Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Ala	Thr	Leu	Ser	Leu	Ser	Pro	Gly
1				5					10					15	

Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Leu	Ser	Tyr	Ile
	20							25				30			

Asn	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Pro	Trp	Ile	Tyr
	35					40					45				

Ala	Thr	Ser	Ser	Leu	Gln	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser
	50				55					60					

Ala	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu
65				70					75					80	

Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln	His	Trp	Ser	Ser	Asp	Pro	Pro	Thr
			85					90						95	

Ile	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys
	100						105		

<210> SEQ ID NO 123

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 123

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Lys	Pro	Gly	Gly
1				5				10						15	

Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Tyr
	20						25					30			

Ala	Val	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
	35					40					45				

Ala	Ser	Ile	Ser	Gly	Gly	Gly	Ile	Tyr	Ile	Tyr	Tyr	Pro	Asp	Ser	Val
	50				55					60					

Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Asn	Ser	Leu	Tyr
65				70					75					80	

Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85					90						95	

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Ala Arg His Gly Gly Ala Gly Tyr Phe Asp Tyr Trp Gly Gln Gly Thr
 100 105 110

Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 124
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 124

Asp Ser Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Gly Ser Glu Ser Val His Ser Tyr
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Val
 35 40 45

Tyr Asn Ala Lys Thr Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Gly Ser Pro Pro
 85 90 95

Trp Thr Phe Gly Gly Thr Lys Val Glu Ile Lys
 100 105

1. An anti-B7H3 monoclonal antibody or an antigen-binding fragment thereof, comprising a heavy chain CDR and a light chain CDR, wherein the heavy chain CDR and the light chain CDR are selected from one or more groups of CDR combinations denoted as a to n:

Combination name	Heavy chain CDR1	Heavy chain CDR2	Heavy chain CDR3	Light chain CDR1	Light chain CDR2	Light chain CDR3
a	SEQ ID NO: 29	SEQ ID NO: 30	SEQ ID NO: 31	SEQ ID NO: 32	SEQ ID NO: 33	SEQ ID NO: 34
b	SEQ ID NO: 35	SEQ ID NO: 36	SEQ ID NO: 37	SEQ ID NO: 38	SEQ ID NO: 39	SEQ ID NO: 40
c	SEQ ID NO: 41	SEQ ID NO: 42	SEQ ID NO: 43	SEQ ID NO: 44	SEQ ID NO: 45	SEQ ID NO: 46
d	SEQ ID NO: 47	SEQ ID NO: 48	SEQ ID NO: 49	SEQ ID NO: 50	SEQ ID NO: 51	SEQ ID NO: 52
e	SEQ ID NO: 53	SEQ ID NO: 54	SEQ ID NO: 55	SEQ ID NO: 56	SEQ ID NO: 57	SEQ ID NO: 58
f	SEQ ID NO: 59	SEQ ID NO: 60	SEQ ID NO: 61	SEQ ID NO: 62	SEQ ID NO: 63	SEQ ID NO: 64
g	SEQ ID NO: 65	SEQ ID NO: 66	SEQ ID NO: 67	SEQ ID NO: 68	SEQ ID NO: 69	SEQ ID NO: 70
h	SEQ ID NO: 71	SEQ ID NO: 72	SEQ ID NO: 73	SEQ ID NO: 74	SEQ ID NO: 75	SEQ ID NO: 76
i	SEQ ID NO: 77	SEQ ID NO: 78	SEQ ID NO: 79	SEQ ID NO: 80	SEQ ID NO: 81	SEQ ID NO: 82
g	SEQ ID NO: 83	SEQ ID NO: 84	SEQ ID NO: 85	SEQ ID NO: 86	SEQ ID NO: 87	SEQ ID NO: 88
k	SEQ ID NO: 89	SEQ ID NO: 90	SEQ ID NO: 91	SEQ ID NO: 92	SEQ ID NO: 93	SEQ ID NO: 94
l	SEQ ID NO: 95	SEQ ID NO: 96	SEQ ID NO: 97	SEQ ID NO: 98	SEQ ID NO: 99	SEQ ID NO: 100

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Combination name	Heavy chain CDR1	Heavy chain CDR2	Heavy chain CDR3	Light chain CDR1	Light chain CDR2	Light chain CDR3
m	SEQ ID NO: 101	SEQ ID NO: 102	SEQ ID NO: 103	SEQ ID NO: 104	SEQ ID NO: 105	SEQ ID NO: 106
n	SEQ ID NO: 107	SEQ ID NO: 108	SEQ ID NO: 109	SEQ ID NO: 110	SEQ ID NO: 111	SEQ ID NO: 112

2. The anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1, further comprising a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region and the light chain variable region are selected from one or more groups of variable region combinations denoted as A to N:

Combination name	Heavy chain variable region	Light chain variable region
A	SEQ ID NO: 1	SEQ ID NO: 2
B	SEQ ID NO: 3	SEQ ID NO: 4
C	SEQ ID NO: 5	SEQ ID NO: 6
D	SEQ ID NO: 7	SEQ ID NO: 8
E	SEQ ID NO: 9	SEQ ID NO: 10
F	SEQ ID NO: 11	SEQ ID NO: 12
G	SEQ ID NO: 13	SEQ ID NO: 14
H	SEQ ID NO: 15	SEQ ID NO: 16
I	SEQ ID NO: 17	SEQ ID NO: 18
J	SEQ ID NO: 19	SEQ ID NO: 20
K	SEQ ID NO: 21	SEQ ID NO: 22
L	SEQ ID NO: 23	SEQ ID NO: 24
M	SEQ ID NO: 25	SEQ ID NO: 26
N	SEQ ID NO: 27	SEQ ID NO: 28

3. The anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 2, wherein the heavy chain variable region and the light chain variable region are humanized.

4. The anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 3, wherein the humanized heavy chain variable region and light chain variable region are selected from one or more groups of variable region combinations denoted as 1 to 6:

Combination name	Heavy chain variable region	Light chain variable region
1	SEQ ID NO: 113	SEQ ID NO: 114
2	SEQ ID NO: 115	SEQ ID NO: 116
3	SEQ ID NO: 117	SEQ ID NO: 118
4	SEQ ID NO: 119	SEQ ID NO: 120
5	SEQ ID NO: 121	SEQ ID NO: 122
6	SEQ ID NO: 123	SEQ ID NO: 124

5. The anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 4, further comprising a constant region of human IgG1.

6. A method for using the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1 for at least one of:

regulating B7H3 activity or B7H3 level,
selectively killing or inhibiting a tumor through toxin conjugation,
mediating antibody-dependent cytotoxicity (ADCC),
blocking an effect of B7H3 in immunosuppression to enhance body immunity,
promoting T lymphocyte, or
improving production of cytokine.

7. A monoclonal antibody conjugate, comprising:

a monoclonal antibody and a conjugating part,
wherein the monoclonal antibody is the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1, and

wherein the conjugating part is selected from one or more at least one of a radionuclide, a medication, a toxin, a cytokine, a cytokine receptor fragment, an enzyme, fluorescein, or biotin.

8. A method for using the monoclonal antibody conjugate according to claim 7 for at least one of:

regulating B7H3 activity or B7H3 level,
selectively killing or inhibiting a tumor through toxin conjugation,
mediating antibody-dependent cytotoxicity (ADCC),
blocking an effect of B7H3 in immunosuppression to enhance body immunity,
promoting T lymphocyte, or
improving production of cytokine.

9. A method for using the monoclonal antibody conjugate according to claim 7 for preventing, treating, and/or adjutantly treating a tumor.

10. An expression vector, comprising a nucleic acid molecule of the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1.

11. A pharmaceutical composition, comprising:
the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1; and
a pharmaceutical carrier.

12. The method of claim 6, wherein the cytokine is IFN- γ in T lymphocyte.

13. The method of claim 8, wherein the cytokine is IFN- γ in T lymphocyte.

* * * * *